

Gate (II.11), phase V3:

The continuum torus package A1 — construction of K_L , trace class, gap, and the periodic Feynman–Kac clause (A1-iv); closure of A2(b)

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Abstract

This document proves Statement A1 of the scoping note in full, including the clause (A1-iv) added by V1: for the $P(\varphi)_2$ interaction on the spatial circle $\mathbb{T}_L$ (quartic with quadratic dictionary terms, any Wick convention of the A4b family), the Hamiltonian K_L is constructed as the strong limit of mode-truncated semigroups; it is self-adjoint, bounded below, acts as $H_0 + V_L$ on bounded smooth cylinders, and its form restricts to the form sum there. e^{-tK_L} is trace class for every $t > 0$ and is the *trace-norm* limit of its cylinder truncations; the spectrum is discrete, obeys $\mu_j \geq \frac{1}{2}\nu_j - c$ against the free eigenvalues, and the truncated eigenvalues converge. The semigroup is positivity improving, so the ground state is simple and strictly positive and the gap $\gamma_L > 0$ is automatic; $\omega_{L,\lambda}^{\text{ct}}$ is its vector state on the Weyl algebra. For $t \geq t_0$ with $t_0 L \geq 1$, the periodic Feynman–Kac identities over $\mathbb{T}_t \times \mathbb{T}_L$ hold: $\text{Tr } e^{-tK_L} = Z_0(t) \mathbb{E}_{\mu^{\text{per}}}[e^{-\mathcal{U}_t}]$, and the Weyl-insertion traces equal the V1 mean-shift functionals — so A2(b) closes as a corollary. The route is the sanctioned from-scratch fallback: an explicit Gaussian-to-Schrödinger unitary, closed finite-dimensional forms, Mehler kernels, bounded-potential Trotter products, cyclic bridge densities, and a direct Cameron–Martin calculation, followed by V1’s thermal machinery, V2’s ensemble estimates, one elementary Trotter instance (bounded perturbations, as in model-sheet (3.16)), and the Simon-book $\Gamma(A)$ cluster (Thms. I.12, I.16, I.17, Cor. I.15) as the only printed imports, with page images in `dossier_images/`.

Contents

1	Statement, conventions, imports	2
1.1	The package to be proved	2
1.2	Conventions	3
1.3	Imports	3
2	The interactions and the three ensembles	4
2.1	Sharp-time interaction on the circle	4
2.2	The three Gaussian ensembles and the uniform Nelson lemma	5
3	The finite-mode systems	8
4	The Hamiltonian K_L (A1-i)	13
5	Trace class and trace-norm cylinder approximation (A1-ii)	15

6	The periodic Feynman–Kac clause (A1-iv) and the closure of A2(b)	17
6.1	Euclidean-side limits	17
6.2	The clause	18
7	Gap, ground state, and the state on the Weyl algebra (A1-iii)	19
8	Acceptance, ledger, imports	20

1 Statement, conventions, imports

1.1 The package to be proved

Fix $L, m > 0$, $\lambda \geq 0$, and dictionary constants $a, b \in \mathbb{R}$ (the A4b convention family). Wherever a lower time-cutoff $t_0 > 0$ appears below we assume, as V2 does, the standing convention

$$t_0 L \geq 1 \quad (\star)$$

— concretely $t_0 := \max(1, 1/L)$ — since *every* V2 result is stated only under it (V2, §1: “assumed throughout this document; every statement below inherits it”), and the one that forces it — Lemma 3.1(ii) — is *false* without it: at $\chi = 0$ its conclusion reads $\Sigma_1 \leq C_2 t L (1 + \log 2)$ while Σ_1 saturates as $t \downarrow 0$, forcing $C_2 \geq 3.7, 37, 3.7 \times 10^2, 3.7 \times 10^3$ at $t = 1, 10^{-1}, 10^{-2}, 10^{-3}$ ($m = L = 1$). This costs nothing: L is fixed throughout Part II and t_0 is at our disposal, and the gate consumes the package only at $t \geq t_* = \max(1, 1/L)$ (V4, Theorem 6.2). Everything below is proved for arbitrary $(a, b) \in \mathbb{R}^2$; *the gate endpoint is* $(a, b) = (0, 0)$, the plain truncation-convention quartic, because by V2, Corollary 5.2 the lattice-vacuum and continuum truncation conventions agree in the limit: the two Wick constants differ by $\delta_\infty = \frac{1}{2\pi} \log \frac{4}{\pi}$, but so do the two fields’ equal-point variances, and the offsets cancel, leaving no counterterm. (An earlier version declared the endpoint $(a, b) = (-6\delta_\infty, 3\delta_\infty^2)$, following the superseded form of V2, Corollary 5.2; see V2, Remark 5.3 and `strategy.md`.) Write $\Gamma_\infty = \frac{\pi}{L} \mathbb{Z}$, $\gamma(k) = (m^2 + k^2)^{1/2}$, and

$$Z_0(t) := \prod_{k \in \Gamma_\infty} (1 - e^{-t\gamma(k)})^{-1} < \infty \quad (t > 0),$$

finite because $\sum_k e^{-t\gamma(k)} < \infty$. The deliverables, stated precisely as Theorems 4.1–6.2 and Corollary 6.3 below, are:

- (A1-i) a self-adjoint, bounded-below realization K_L of $H_{0,L} + \lambda \int_{\mathbb{T}_L} (:\varphi^4:_{C_L} + a:\varphi^2:_{C_L} + b) dx$, acting as the operator sum on bounded smooth cylinder vectors, with quadratic form equal to the form sum there;
- (A1-ii) e^{-tK_L} trace class for all $t > 0$, with discrete spectrum $\mu_0 < \mu_1 \leq \dots \rightarrow \infty$ and free comparison $\mu_j \geq \frac{1}{2}\nu_j - c$;
- (A1-iii) positivity improving semigroup, hence simple ground state $\Omega_L > 0$ a.e. and gap $\gamma_L := \mu_1 - \mu_0 > 0$; $\omega_{L,\lambda}^{\text{ct}} := \langle \Omega_L, \cdot \Omega_L \rangle$ on the Weyl algebra $\mathfrak{W}(\mathbb{T}_L)$;
- (A1-iv) the periodic Feynman–Kac representation over $\mathbb{T}_t \times \mathbb{T}_L$, compatible with cylinder approximation: the partition identity $\text{Tr} e^{-tK_L} = Z_0(t) \mathbb{E}_{\mu_{t,L}^{\text{per}}}[e^{-\mathcal{U}_t}]$, the trace-norm convergence of cylinder truncations, and the Weyl-insertion identity in the exact mean-shift form of V1, Definition 8.1 — which is Statement A2(b) (Corollary 6.3).

1.2 Conventions

$\mathbb{T}_L = \mathbb{R}/2L\mathbb{Z}$; real orthonormal γ -eigenbasis of $L^2(\mathbb{T}_L)$:

$$e_0 := (2L)^{-1/2}, \quad e_k^c := L^{-1/2} \cos(kx), \quad e_k^s := L^{-1/2} \sin(kx) \quad (k \in \Gamma_\infty, k > 0),$$

each an eigenfunction of $\gamma = (m^2 - \partial_x^2)^{1/2}$ with eigenvalue $\gamma(k)$. Enumerate this basis as $(e_j)_{j \geq 1}$ with dispersions (γ_j) , each $\gamma_j \geq m$, and let

$$(Q, \mu) := \left(\mathbb{R}^\infty, \bigotimes_{j \geq 1} \mathcal{N}(0, 1) \right), \quad \xi = (\xi_j)_{j \geq 1},$$

the standard Gaussian product space. The time-zero field and its truncations are

$$\varphi(x) := \sum_j \frac{\xi_j}{(2\gamma_j)^{1/2}} e_j(x), \quad \varphi^{(K)}(x) := \sum_{j: |k_j| \leq K} \frac{\xi_j}{(2\gamma_j)^{1/2}} e_j(x), \quad (1)$$

the first as an $L^2(\mu)$ -valued distribution, the second an a.s. smooth random field in finitely many coordinates (n_K of them). Since $\sum_{|k_j| \leq K} e_j(x)^2 = (2L)^{-1} \# \{k \in \Gamma_\infty : |k| \leq K\}$ pointwise (the $\cos^2 + \sin^2$ pairing), the truncated equal-point variance is x -independent:

$$c_K := \mathbb{E}_\mu[\varphi^{(K)}(x)^2] = \frac{1}{2L} \sum_{k \in \Gamma_\infty, |k| \leq K} \frac{1}{2\gamma(k)} \leq C_c(m, L) (1 + \log(1 + K/m)), \quad (2)$$

by the integral comparison $\sum_{|k| \leq K} \gamma(k)^{-1} \leq \frac{2L}{\pi} \int_0^K (m^2 + u^2)^{-1/2} du + m^{-1}$. This c_K is the Γ -truncated continuum vacuum constant $c_{L,\cdot}^{\text{ct}}$ of V2, §5. On $L^2(Q, \mu)$ the free Hamiltonian is $H_0 := d\Gamma(\gamma) \geq 0$, with $e^{-tH_0} = \Gamma(e^{-t\gamma})$, $\text{Tr } e^{-tH_0} = Z_0(t)$, and eigenvalues $\nu_0 = 0 < \nu_1 \leq \dots$ (occupation sums $\sum n_j \gamma_j$). The *cylinder core* is

$$\mathcal{D} := \{\psi(\xi) = F(\xi_1, \dots, \xi_n) : n \in \mathbb{N}, F \in C_b^\infty(\mathbb{R}^n)\},$$

dense in $L^2(\mu)$, contained in $D(H_0)$, with $H_0\psi = \sum_j \gamma_j(-\partial_j^2 + \xi_j \partial_j)\psi$ there.

1.3 Imports

Import 1.1 (the $\Gamma(A)$ cluster; Simon, *The $P(\phi)_2$ Euclidean (Quantum) Field Theory*). For a contraction A of real Hilbert spaces and the second quantization $\Gamma(A)$ between the associated Gaussian L^2 -spaces: (a) $\Gamma(A)$ is positivity preserving [Thm. I.12; dossier image p.48]; (b) $\Gamma(A)$ is a contraction on each L^p , $1 \leq p \leq \infty$ [Cor. I.15; p.53]; (c) if $\|A\| < 1$ then $\Gamma(A)$ is positivity improving [Thm. I.16; p.55]; (d) $\|\Gamma(A)\psi\|_{L^q} \leq \|\psi\|_{L^p}$ whenever $1 < p \leq q < \infty$ and $\|A\| \leq (p-1)^{1/2}(q-1)^{-1/2}$ [Thm. I.17 (Nelson); p.56 = printed p.34]. This is the identical import cluster as Part I's Assumption **A:hyper** (same source, same theorem I.17); the abstract Gaussian form applies verbatim to every Gaussian space used below (vacuum, periodic, Ornstein–Uhlenbeck). Page images: dossier_images/simon-gamma-p{48,53,55,56,57}.png.

Import 1.2 (program-internal). (a) Model sheet §3 (`limport_part_ii_model_sheet`): the finite-dimensional pattern — coercive closed form, compact resolvent, Feynman–Kac kernel (3.16) via Trotter for bounded truncations plus monotone form convergence, positivity improving, simple ground state. The same elementary Trotter instance (self-adjoint semi-bounded + bounded perturbation) is invoked below exactly as there. (b) V1 (`v1_thermal_representations`): Propositions 2.1, 3.1, Lemma 3.2, Theorem 3.3 (kernel identity), Lemmas 4.1–4.2, Theorem 4.3 and Theorem 6.1 (free closed forms, lattice and continuum),

Lemmas 5.1–5.3, Theorem 5.4 (mean shift), Remark 5.5 (ensemble), Definition 8.1 and Remark 8.2 (the A2(b) form and the (A1-iv) clause). (c) V2 (`v2_interaction_comparison`): §1 (periodic ensemble $\mu_{t,L}^{\text{per}}$, Prop. 1.2), §2 (recombination Lemma 2.1; existence of the continuum chaoses, Prop. 2.4), §3 (envelope Lemma 3.1, symbol bounds), §5 (dictionary), §6 (closed-form interpolant Lemma 6.1, shift fields), §7 (uniform Nelson bounds, Theorem 7.2 and its proof pattern). (d) Part I Appendix C: the $\Gamma(e^{-\tau})$ sum-of-chaoses moment comparison and the layer-cake assembly (proof of `thm:stability`), as already re-used in V2 §7.

Chaos moment bound (consequence of Import 1.1(d), used throughout): if X is a sum of Wick chaoses of order ≤ 4 in any Gaussian space, then $\|X\|_{L^q} \leq (q-1)^2 \|X\|_{L^2}$ for $q \geq 2$ (Part I, App. C form: write $X = \Gamma(e^{-\tau}) \sum_r e^{r\tau} X_r$ and use orthogonality; $q = 1 + e^{2\tau}$).

2 The interactions and the three ensembles

2.1 Sharp-time interaction on the circle

Definition 2.1. For $K \in \Gamma_\infty$, $K > 0$, set (all Wick powers with the constant (2))

$$V^{(K)} := \lambda \int_{\mathbb{T}_L} \left[: \varphi^{(K)}(x)^4 :_{c_K} + a : \varphi^{(K)}(x)^2 :_{c_K} + b \right] dx,$$

a polynomial in the n_K variables $(\xi_j)_{|k_j| \leq K}$.

Lemma 2.2 (pointwise bound and coercivity). *For every $u \in \mathbb{R}$ and $c \geq 0$,*

$$:u^4:_c + a :u^2:_c + b = (u^2 - 3c)^2 + a(u^2 - 3c) - 6c^2 + 2ac + b \geq -\left(6c^2 + 2|a|c + \frac{a^2}{4} + |b|\right). \quad (3)$$

Hence $V^{(K)} \geq -2L\lambda(6c_K^2 + 2|a|c_K + \frac{a^2}{4} + |b|) =: -b_K$, with $b_K \leq c_1(1 + \log(1 + K/m))^2$, $c_1 = c_1(\lambda, a, b, m, L)$. Moreover, for $\lambda > 0$,

$$V^{(K)} \geq \frac{\lambda}{2L} \left(\int_{\mathbb{T}_L} (\varphi^{(K)})^2 dx \right)^2 - C_K \left(1 + \int_{\mathbb{T}_L} (\varphi^{(K)})^2 dx \right), \quad \int_{\mathbb{T}_L} (\varphi^{(K)})^2 dx = \sum_{|k_j| \leq K} \frac{\xi_j^2}{2\gamma_j},$$

with a finite constant C_K ; in particular $V^{(K)}(\xi) \rightarrow +\infty$ as $|\xi| \rightarrow \infty$ in $\mathbb{R}^{n_K}$ (for $\lambda = 0$ the total potential below is purely harmonic).

Proof. (3): substitute $v := u^2 - 3c \geq -3c$ and minimize $v^2 + av \geq -a^2/4$; the constant collects as displayed. Integrate over $\mathbb{T}_L$ for the first claim, insert (2) for b_K . Coercivity: $\int (\varphi^{(K)})^4 \geq (2L)^{-1} (\int (\varphi^{(K)})^2)^2$ by Cauchy–Schwarz on $\mathbb{T}_L$, and the Wick-subtracted quadratic terms are linear in $\int (\varphi^{(K)})^2$ with K -dependent coefficients; Parseval gives the mode expression. $\square$

Lemma 2.3 (L^2 structure and rates, vacuum ensemble). *Each $V^{(K)}$ is a sum of Wick chaoses of order ≤ 4 in $L^2(\mu)$, $\mathbb{E}_\mu[V^{(K)}] = 2L\lambda b$, and there is $V_L \in L^2(\mu)$ (chaos ≤ 4) with, for $K' \geq K \geq m$,*

$$\|V^{(K)} - V^{(K')}\|_{L^2(\mu)} + \|V^{(K)} - V_L\|_{L^2(\mu)} \leq C_V \frac{1 + \log(1 + K/m)}{(K/m)^{1/2}}, \quad C_V = C_V(\lambda, a, m, L). \quad (4)$$

Consequently $V^{(K)} \rightarrow V_L$ in every $L^p(\mu)$, $p < \infty$ (chaos moment bound), and $V_L = \lambda \int_{\mathbb{T}_L} (: \varphi^4 :_{c_L} + a : \varphi^2 :_{c_L} + b) dx$ in the sense of the truncation convention (V2, §5).

Proof. The quartic chaos norm: by V2's Isserlis Lemma 2.3 applied at sharp time, with the one-dimensional momentum-space covariance symbol $\hat{C}(k) = (2\gamma(k))^{-1}$,

$$\left\| \int_{\mathbb{T}_L} :(\varphi^{(K)})^4: - :(\varphi^{(K')})^4: dx \right\|_2^2 = 4! (2L)^{-2} \sum_{\substack{k_1 + \dots + k_4 = 0 \\ \text{some } |k_i| \in (K, K']}} \prod_{i=1}^4 \frac{1}{2\gamma(k_i)},$$

i.e. $4! (2L)^{-2}$ times the constrained sum of $\prod (2\gamma(k_i))^{-1}$ (the bookkeeping is V2, Lemma 4.1 with the time direction absent). Dominate by the one-dimensional convolution decay: with $g(k) := (m + |k|)^{-1} \geq (2\gamma)^{-1} \cdot \text{const}^{-1}$ reversed, i.e. $(2\gamma(k))^{-1} \leq g(k)$, the three-region argument of V2, Lemma 3.1(ii) (one dimension, sums over Γ_∞ with density L/π) gives

$$(g * g)(k) \leq CL \frac{1 + \log(2 + |k|/m)}{m + |k|}, \quad (g * g * g)(k) \leq CL^2 \frac{(1 + \log(2 + |k|/m))^2}{m + |k|},$$

so the tail sum with $|k_1| > K$ is bounded by $\sum_{|k_1| > K} g(k_1) \cdot CL^2 \log^2(2 + |k_1|/m) (m + |k_1|)^{-1} \leq CL^3 \log^2(1 + K/m) (m/K)$.

The chaos structure is nested with *no* cross-convention terms: since c_K is the variance of $\varphi^{(K)}(x)$ itself, $:\varphi^{(K)}(x)^4:_{c_K}$ is the pure fourth Wiener chaos of $\varphi^{(K)}(x)^4$ and $:\varphi^{(K)}(x)^2:_{c_K}$ the pure second chaos (own-variance Wick ordering is the Hermite/chaos projection), so

$$V^{(K)} = \lambda I_4(f_K) + \lambda a I_2(g_K) + 2L\lambda b$$

exactly, with mode kernels f_K, g_K equal to the restrictions of the fixed kernels f_∞, g_∞ to modes $|k_i| \leq K$. Hence $\mathbb{E}_\mu[V^{(K)}] = 2L\lambda b$ for every K , and $V^{(K')} - V^{(K)}$ is exactly the chaos integral of the kernel tails — the constrained sums displayed above for the quartic part, and for the quadratic part the diagonal tail: that difference is $\lambda a \sum_{K < |k_j| \leq K'} (\xi_j^2 - 1)/(2\gamma_j)$ over the tail modes, of variance $\frac{1}{2} \lambda^2 a^2 \sum_{K < |k| \leq K'} \gamma(k)^{-2} \leq C(a, \lambda) L/K$ (the k -sum is $\leq \frac{2L}{\pi} \int_K^\infty (m^2 + u^2)^{-1} du \leq 2L/(\pi K)$). Taking square roots gives (4); completeness of L^2 gives $V_L = \lambda I_4(f_\infty) + \lambda a I_2(g_\infty) + 2L\lambda b$. The identification with the truncation convention is V2, §5 (the proof of its Corollary 5.2, first paragraph). $\square$

2.2 The three Gaussian ensembles and the uniform Nelson lemma

Three Gaussian ensembles carry the analysis; all chaos and moment statements below use the abstract Import 1.1(d).

Definition 2.4 (ensembles). (i) *Vacuum*: (Q, μ) as above. (ii) *Periodic*: $\mu_{t,L}^{\text{per}}$, the V2 ensemble on $\mathbb{T}_t \times \mathbb{T}_L$ (V2, Definition 1.1 and Prop. 1.2): mode covariances $c_\omega(s - s') = \cosh(\omega(\frac{t}{2} - |s - s'|)) / (2\omega \sinh(\frac{\omega t}{2}))$ at $\omega = \gamma(k)$. (iii) *Ornstein–Uhlenbeck*: the stationary field process $(X_s)_{s \in \mathbb{R}}$ with time-zero marginal μ and per-mode covariance $\mathbb{E}[\xi_j(s) \xi_j(s')] = e^{-\gamma_j |s - s'|}$; concretely, independent one-dimensional stationary OU processes in each coordinate, with continuous versions (Kolmogorov–Chentsov, as the covariance is Lipschitz). For s, t fixed, the pair (X_0, X_t) is Gaussian with per-mode correlation $e^{-\gamma_j t}$; equivalently, for bounded F, G ,

$$\mathbb{E}[F(X_0) G(X_t)] = \langle \bar{F}, \Gamma(e^{-t\gamma}) G \rangle_{L^2(\mu)}. \quad (5)$$

(5) holds because both sides are Gaussian expectations with the same finite-dimensional covariances (per mode, the Mehler operator's kernel is the two-point OU density; cylinder functions, then L^2 limits).

Lemma 2.5 (abstract uniform Nelson bound). *Let $(\Omega, \mathbb{P})$ be a Gaussian space and $\{A_K\}_{K \in \mathcal{K}}$, $\mathcal{K} \subseteq \Gamma_\infty$ unbounded, a family of sums of Wick chaoses of order ≤ 4 with, for constants $c_1, C_A, \kappa > 0$: (N0) consecutive elements of $\mathcal{K}$ satisfy $\ell_{K^+} - \ell_K \leq \log 2$, where $\ell_K := 1 + \log(1 + K/m)$ and K^+ is the successor of K in $\mathcal{K}$; (N1) $A_K \geq -c_1(1 + \log(1 + K/m))^2$ a.s.; (N2) $\|A_K - A_{K'}\|_{L^2(\mathbb{P})} \leq C_A(1 + \log(1 + K/m))(K/m)^{-\kappa}$ for $K' \geq K$. Then for every $p < \infty$,*

$$\sup_{K \in \mathcal{K}} \|e^{-A_K}\|_{L^p(\mathbb{P})} \leq B(p, c_1, C_A, \kappa, m, \ell_{K_0}) < \infty, \quad K_0 := \min \mathcal{K},$$

and if $A_K \rightarrow A$ in L^2 then $e^{-A} \in L^p(\mathbb{P})$ with the same bound. The dependence on $\ell_{K_0} = 1 + \log(1 + K_0/m)$ cannot be dropped: the constant family $A_K := -c_1 \ell_{K_0}^2$ on $\mathcal{K} = \Gamma_\infty \cap [K_0, \infty)$ satisfies (N0)–(N2) at fixed (p, c_1, C_A, κ, m) — a constant is a chaos of order 0, (N1) holds with equality at $K = K_0$, and (N2) holds with left-hand side 0 (the right-hand side being $C_A \ell_K (K/m)^{-\kappa} > 0$, since $C_A > 0$) — yet $\sup_K \|e^{-A_K}\|_{L^p} = e^{c_1 \ell_{K_0}^2} \rightarrow \infty$ as $K_0 \rightarrow \infty$. In every instantiation below $\mathcal{K}$ is a full tail of Γ_∞ with a fixed base point K_0 , so the resulting bounds are K -uniform as claimed.

Proof. This is the proof of V2, Theorem 7.2, with the dyadic scale variable $\log(1/\varepsilon)$ replaced by $\ell_K := 1 + \log(1 + K/m)$; we record the instance. For b with $L_b := (b/2c_1)^{1/2} \geq \ell_{K_0} \vee 2 \log 2$ ($K_0 := \min \mathcal{K}$) choose the largest $K_b \in \mathcal{K} \cap [K_0, K]$ with $\ell_{K_b} \leq L_b$; by (N0) this satisfies $\ell_{K_b} > L_b - \log 2 \geq L_b/2$ when $b \leq b_K^* := c_1 \ell_K^2$ (indeed $\ell_{K_b^+} > L_b$ by maximality and $\ell_{K_b} \geq \ell_{K_b^+} - \log 2$; for $b > b_K^*$, $\mathbb{P}[A_K \leq -b] = 0$ by (N1)). Hypothesis (N0) is not decorative — it is exactly what the scale selection uses, and without it *this proof's estimate* fails. Take the tower family $K_0 := \pi/L$, $K_{n+1} := (\pi/L) 2^{(L/\pi)K_n}$ (which lies in Γ_∞ , unlike a naive e^{K_n}), for which $\ell_{K_{n+1}} = 1 + \log(1 + \frac{\pi}{Lm} 2^{(L/\pi)K_n}) \sim (\log 2)(L/\pi)K_n$ (the factor $\pi/(Lm)$ comes from $\ell_K = 1 + \log(1 + K/m)$; it shifts the value by the additive constant $\log(\pi/(Lm))$ and leaves the asymptotics unchanged). For L_b falling in such a gap one has only $\ell_{K_b} = O(\log L_b)$, whence by (N2)

$$\rho_b \leq C_A \ell_{K_b} (K_b/m)^{-\kappa} \sim C_A L_b^{-\kappa} \log L_b,$$

merely polynomially rather than exponentially small in L_b ; optimizing Chebyshev then gives a tail exponent $\exp(-cL_b^{1+\kappa/2}(\log L_b)^{-1/2})$ — a fortiori no better than $\exp(-cL_b^{1+\kappa/2})$ — and since $b = 2c_1 L_b^2$, for the $\kappa \leq 1$ of every instantiation below ($\kappa = \frac{1}{2}$, resp. $\frac{1}{2} - \varsigma$) the exponent $1 + \kappa/2 \leq \frac{3}{2} < 2$ is too small to beat pe^{pb} : the layer-cake bound diverges and the argument breaks down. (This exhibits the failure of the proof's own estimate, not of the conclusion: (N1)–(N2) are upper bounds, so no divergence of the true $\mathbb{E}[e^{-pA_K}]$ follows.) Divergent ℓ -gaps alone do *not* suffice: for the factorial family $K_n = (\pi/L)[e^{n!}]$ one still has $\ell_{K_b} \geq L_b/(n+1)$, i.e. $\ell_{K_b} \geq c_b L_b$ with $c_b \rightarrow 0$ only logarithmically in L_b , and the proof still closes, with κ replaced by $c_b \kappa$ at each b ; the witness must be at least tower-sparse. Then $A_{K_b} \geq -c_1 L_b^2 = -b/2$ a.s. and $\{A_K \leq -b\} \subseteq \{A_K - A_{K_b} \leq -b/2\}$, with $\rho_b := \|A_K - A_{K_b}\|_2 \leq 2C_A L_b e^{-\kappa(L_b/2-1)}$ by (N2) and $\ell_{K_b} \geq L_b/2$. The chaos moment bound and Chebyshev at $(2q)^2 = e^{\kappa L_b/4}$ give $\mathbb{P}[A_K \leq -b] \leq \exp(-e^{\kappa L_b/8})$ for $b \geq b_1(c_1, C_A, \kappa)$, uniformly in K ; the layer-cake integral closes as in V2/Part I App. C. The final claim follows since $e^{-A_K} \rightarrow e^{-A}$ in probability along a subsequence and Fatou applies. $\square$

Proposition 2.6 (the three instantiations). *For every $p < \infty$ and $0 < t \leq T$: (i) $\sup_K \|e^{-\sigma V^{(K)}}\|_{L^p(\mu)} < \infty$ for each $\sigma \in (0, \sigma_0]$, any fixed σ_0 , and $e^{-\sigma V_L} \in L^p(\mu)$; (ii) with $\mathcal{W}_K := \int_0^t V^{(K)}(X_s) ds$ on the OU ensemble: $\sup_K \|e^{-\mathcal{W}_K}\|_{L^p} < \infty$ uniformly in $t \leq T$, $\mathcal{W}_K \rightarrow \mathcal{W} := \int_0^t V_L(X_s) ds$ in L^2 with $\|\mathcal{W}_K - \mathcal{W}\|_2 \leq t \|V^{(K)} - V_L\|_{L^2(\mu)}$, and $e^{-\mathcal{W}} \in L^p$; (iii) with $\mathcal{U}_t^{(K)}$ the K -mode truncation of the V2 interaction $\mathcal{U}_t$ on $\mu_{t,L}^{\text{per}}$ (V2, Definition 2.2 with*

all spatial modes restricted to $|k| \leq K$; V2's $\mathcal{U}_t$ is the $(a, b) = (0, 0)$ member of the family, the (a, b) -terms of Definition 2.1 being carried along additively): $\sup_K \|e^{-\mathcal{U}_t^{(K)}}\|_{L^p} < \infty$ for $t \in [t_0, T]$, $\mathcal{U}_t^{(K)} \rightarrow \mathcal{U}_t$ in $L^2(\mu^{\text{per}})$, and $e^{-\mathcal{U}_t} \in L^p$.

Proof. In each case verify (N1), (N2) of Lemma 2.5. In all three cases $\mathcal{K}$ is a full tail $\Gamma_\infty \cap [K_0, \infty)$, for which (N0) holds automatically: for $K \in \Gamma_\infty$, $K \geq \pi/L$, the successor is $K^+ = K + \pi/L$ and $\ell_{K^+} - \ell_K = \log(1 + \frac{\pi/L}{m+K}) \leq \log(1 + \frac{\pi/L}{m+\pi/L}) < \log 2$. (i) (N1) is Lemma 2.2 (scaled by σ); (N2) is (4) with $\kappa = 1/2$. (ii) (N1): pointwise on paths, $\mathcal{W}_K \geq -tb_K \geq -Tc_1\ell_K^2$; (N2): by stationarity and Cauchy–Schwarz in s ,

$$\|\mathcal{W}_K - \mathcal{W}_{K'}\|_{L^2}^2 = \int_0^t \int_0^t \mathbb{E}[\Delta V(X_s) \Delta V(X_{s'})] ds ds' \leq t^2 \|\Delta V\|_{L^2(\mu)}^2,$$

since each two-time expectation is bounded by the equal-time one (Cauchy–Schwarz plus stationarity); then (4). The chaos structure: $\mathcal{W}_K$ is a time integral of chaoses of order ≤ 4 of the global Gaussian field, hence itself chaos ≤ 4 . The limit $\mathcal{W}$ is defined as the L^2 limit, and $e^{-\mathcal{W}} \in L^p$ by the final clause. (iii) Pointwise on paths, $\mathcal{U}_t^{(K)} = \lambda \int_0^t \int_{\mathbb{T}_L} (:X_s^4 :_{c_K} + a :X_s^2 :_{c_K} + b)$, so Lemma 2.2, with the truncated *vacuum* constant c_K , gives $\mathcal{U}_t^{(K)} \geq -c'_1 t \ell_K^2$. We now verify (N2) explicitly.

Let $Y_j^{(K)}$ be the periodic spacetime integral of the j th chaos with all spatial input modes restricted to $|k| \leq K$. Wick isometry and the continuum covariance envelope give, for $K' \geq K \geq m$,

$$\|Y_2^{(K')} - Y_2^{(K)}\|_2^2 \leq C \sum_{\substack{|k| > K \\ k_0 \in (2\pi/t)\mathbb{Z}}} (m^2 + k_0^2 + k^2)^{-2} \leq C(m, L, t_0)(1+t)K^{-2}. \quad (6)$$

Indeed, with $A^2 = m^2 + k^2$,

$$\sum_{k_0 \in (2\pi/t)\mathbb{Z}} (A^2 + k_0^2)^{-2} \leq A^{-4} + \frac{t}{2\pi} \int_{\mathbb{R}} (A^2 + u^2)^{-2} du = A^{-4} + \frac{t}{4A^3},$$

and the spatial sum–integral comparisons give $\sum_{|k| > K} A^{-3} \leq CLK^{-2}$ and $\sum_{|k| > K} A^{-4} \leq CLK^{-3}$.

For the fourth chaos, choose one of the four inputs whose spatial momentum has modulus $> K$ and sum the other three constrained inputs by the V2 convolution bound. Including the factor 4 for this union and the harmless Wick and volume normalizations,

$$\begin{aligned} \|Y_4^{(K')} - Y_4^{(K)}\|_2^2 &\leq C(m, t_0) \sum_{\substack{|k| > K \\ k_0 \in (2\pi/t)\mathbb{Z}}} \frac{[1 + \log(2 + |(k_0, k)|)]^2}{(m^2 + k_0^2 + k^2)^2} \\ &\leq C(m, L, t_0)(1+t)K^{-2}[1 + \log(1 + K/m)]^2. \end{aligned} \quad (7)$$

For the second inequality, the same temporal sum–integral comparison, after splitting at $|k_0| = (m^2 + k^2)^{1/2}$, gives

$$\sum_{k_0} \frac{[1 + \log(2 + |(k_0, k)|)]^2}{(m^2 + k_0^2 + k^2)^2} \leq C(m, t_0)(1+t) \frac{[1 + \log(2 + |k|)]^2}{(m^2 + k^2)^{3/2}}.$$

The remaining decreasing tail is bounded by its first term plus $\frac{L}{\pi} \int_K^\infty (1 + \log(2 + u))^2 (m^2 + u^2)^{-3/2} du$, which is at most $CLK^{-2}[1 + \log(1 + K/m)]^2$ by one integration by parts.

The exact thermal-to-vacuum Wick coefficient is

$$d^{(K)}(t) = \frac{1}{2L} \sum_{|k| \leq K} \frac{1}{\gamma(k)(e^{t\gamma(k)} - 1)}.$$

Since $t \geq t_0$ and $\gamma(k) = \sqrt{m^2 + k^2} \geq \max(m, |k|)$,

$$|d^{(K')}(t) - d^{(K)}(t)| \leq C(m, L, t_0) \sum_{|k| > K} \frac{e^{-t_0\gamma(k)}}{\gamma(k)} \leq C(m, L, t_0)e^{-t_0K/2}. \quad (8)$$

The ensemble-Wick identity is, exactly,

$$\mathcal{U}_t^{(K)} = \lambda[Y_4^{(K)} + (6d^{(K)} + a)Y_2^{(K)} + (3(d^{(K)})^2 + ad^{(K)} + b)(2Lt)]. \quad (9)$$

The norms $\|Y_2^{(K)}\|_2$ and $|d^{(K)}|$ are uniformly bounded by the convergent full sums. Subtracting (9) at K, K' , applying (6)–(8), and taking square roots gives the stronger increment estimate

$$\|\mathcal{U}_t^{(K')} - \mathcal{U}_t^{(K)}\|_2 \leq C(m, L, t_0, T, \lambda, a, b) \ell_K(K/m)^{-1}. \quad (10)$$

Thus (N2) holds directly with $\kappa = 1$; no unspecified “same envelope count” and no logarithm absorption are used. Lemma 2.5 now proves the asserted periodic L^p bounds and the L^2 limit. $\square$

Corollary 2.7 (uniform lower bounds). *There are $c_\lambda = c(\lambda, a, b, m, L, t_0)$ and, for each $\delta \in (0, 1)$, c_δ (the value of $c_\lambda/(1-\delta)$ scaled by $1 - \delta$) such that the finite-mode operators of §3 obey, uniformly in K ,*

$$\inf \text{spec } K^{(K)} \geq -c_\lambda, \quad K^{(K)} \geq \delta H_0 - c_\delta \quad (\text{quadratic forms}).$$

Proof. By the trace identity of Theorem 3.1(v) below (whose proof does not use this corollary), $e^{-t_0} \inf \text{spec } K^{(K)} \leq \text{Tr } e^{-t_0 K^{(K)}} = Z_0(t_0) \mathbb{E}[e^{-\mathcal{U}_{t_0}^{(K)}}] \leq Z_0(t_0)B$ with B from Proposition 2.6(iii), giving the first bound. For the second: $K^{(K)} - \delta H_0 = (1 - \delta)[H_0 + (1 - \delta)^{-1}V^{(K)}]$, and $(1 - \delta)^{-1}V^{(K)}$ is the interaction with parameters (λ, a, b) replaced by $(\lambda/(1 - \delta), a, b)$, so the first bound applies at the amplified coupling. $\square$

3 The finite-mode systems

Fix K . The truncated system is the quantum mechanics of the n_K oscillator modes $\{j : |k_j| \leq K\}$ with dispersions γ_j and the interaction $V^{(K)}$, tensored with the free remainder: on $L^2(\mu) = L^2(\mu_{\leq K}) \otimes L^2(\mu_{>K})$ set

$$K^{(K)} := (H_0^{\leq K} + V^{(K)}) \otimes \mathbf{1} + \mathbf{1} \otimes H_0^{>K}, \quad T_K(t) := e^{-tK^{(K)}}, \quad (11)$$

where $H_0^{\leq K} + V^{(K)}$ is the closed-form realization constructed in Theorem 3.1(i) below.

Theorem 3.1 (finite-mode package in explicit Gaussian coordinates). *For every K and $t > 0$: (i) the form sum $H_0^{\leq K} + V^{(K)}$ is a lower-bounded closed quadratic form on $L^2(\mu_{\leq K})$ whose operator has compact resolvent and simple lowest eigenvalue; $K^{(K)}$ is self-adjoint and bounded below. (ii) $e^{-t(H_0^{\leq K} + V^{(K)})}$ has a jointly continuous, strictly positive integral kernel with respect to $\mu_{\leq K}$, given by the Feynman–Kac formula with respect to the n_K -mode OU/Mehler reference; matrix elements obey, for bounded measurable f, g on $\mathbb{R}^{n_K}$,*

$$\langle f, e^{-t(H_0^{\leq K} + V^{(K)})} g \rangle_{L^2(\mu_{\leq K})} = \mathbb{E} \left[\overline{f(X_0^{\leq K})} g(X_t^{\leq K}) e^{-\int_0^t V^{(K)}(X_s) ds} \right], \quad (12)$$

the expectation over the OU ensemble of Definition 2.4 (only the $\leq K$ coordinates enter the integrand). (iii) Consequently, for bounded measurable cylinder f, g (any finitely many modes), $\langle f, T_K(t)g \rangle = \mathbb{E}[\bar{f}(X_0)g(X_t)e^{-\int_0^t V^{(K)}(X_s)ds}]$. (iv) $T_K(t)$ is trace class, with $\text{Tr } T_K(t) \leq e^{tb_K} Z_0(t)$ (b_K from Lemma 2.2). (v) The periodic trace identity holds:

$$\text{Tr } T_K(t) = Z_0(t) \mathbb{E}_{\mu_{t,L}^{\text{per}}} [e^{-\mathcal{U}_t^{(K)}}].$$

(vi) For every Weyl symbol $\xi = \varepsilon_M q + ip$ whose refined momentum and position parts are supported in modes $|k| \leq K$, the following mode-by-mode mean-shift identity holds with dispersion $\gamma(\cdot)$:

$$\frac{\text{Tr}(T_K(t) W_{\text{ct}}(\xi))}{\text{Tr } T_K(t)} = e^{-S^{(K)}(H_p)} \frac{\mathbb{E}_{\mu^{\text{per}}} [e^{i\Phi(J_q)} e^{-\mathcal{U}_t^{(K)}(\eta + H_p^{(K)})}]}{\mathbb{E}_{\mu^{\text{per}}} [e^{-\mathcal{U}_t^{(K)}}]}, \quad (13)$$

with the γ -dispersion action in the normalization of V1, eq. (12) and V4, eq. (3),

$$S^{(K)}(H_p) = \frac{1}{4 \cdot 2L} \sum_{|k| \leq K} \gamma(k) \coth\left(\frac{t\gamma(k)}{2}\right) |b_\infty(k)|^2, \quad b_\infty(k) := \varepsilon_M^{1/2} P_\infty(\varepsilon_M k) \hat{p}_M(k),$$

$H_p^{(K)}$ the interpolant with the closed form of V2, Lemma 6.1 at $\omega = \gamma(k)$, and J_q the sharp-time position source.

Proof. Let J_K be the finite set of retained real sine/cosine coordinates and $n_K = |J_K|$. The relation to the ordinary oscillator Schr"odinger coordinates is fixed, not implicit: for one mode define

$$(\mathcal{U}_j f)(q) := \left(\frac{\gamma_j}{\pi}\right)^{1/4} e^{-\gamma_j q^2/2} f(\sqrt{2\gamma_j} q), \quad (14)$$

and let $\mathcal{U}_K = \bigotimes_{j \in J_K} \mathcal{U}_j$. With $d\mu_1(\xi) = (2\pi)^{-1/2} e^{-\xi^2/2} d\xi$, the substitution $\xi = \sqrt{2\gamma_j} q$ gives

$$\int_{\mathbb{R}} |\mathcal{U}_j f(q)|^2 dq = \int_{\mathbb{R}} |f(\xi)|^2 d\mu_1(\xi),$$

so (14) is unitary. Direct differentiation yields

$$\mathcal{U}_j \gamma_j (-\partial_\xi^2 + \xi \partial_\xi) \mathcal{U}_j^{-1} = \frac{1}{2} (-\partial_q^2 + \gamma_j^2 q^2 - \gamma_j). \quad (15)$$

Moreover $q_j = \xi_j / \sqrt{2\gamma_j}$, exactly the coefficient in the field formula (1). Thus the interaction polynomial is the same function on both sides of the unitary; no scaling constant is suppressed.

Choose $C > b_K + 1$ and put $W = V^{(K)}$. In Gaussian coordinates the form is

$$q_K(\psi) = \sum_{j \in J_K} \gamma_j \|\partial_j \psi\|_{L^2(\mu)}^2 + \int (W + C) |\psi|^2 d\mu. \quad (16)$$

It is dense because $C_c^\infty(\mathbb{R}^{n_K})$ is contained in its domain. If (ψ_n) is Cauchy in $\|\cdot\|_2^2 + q_K$, then it converges in $L^2(\mu)$ and each weighted weak derivative converges in $L^2(\mu)$. Choose a subsequence converging pointwise a.e.; closedness of weak derivatives identifies their limits with the derivatives of ψ , while Fatou gives $\int (W + C) |\psi|^2 d\mu \leq \liminf_n \int (W + C) |\psi_n|^2 d\mu$. Applying the same argument to $\psi_n - \psi_m$ and then $m \rightarrow \infty$ shows convergence in the form norm. Hence (16) is closed and lower bounded. This is the form of $H_0^{\leq K} + W + C$; subtracting the scalar C gives the asserted realization of $H_0^{\leq K} + W$.

Compactness must also cover $\lambda = 0$, so it cannot be inferred from coercivity of W . Instead use the free form. Since $W \geq -b_K$ and $C > b_K + 1$,

$$q_K(\psi) \geq q_{0,K}(\psi) + \|\psi\|_2^2, \quad q_{0,K}(\psi) := \sum_{j \in J_K} \gamma_j \|\partial_j \psi\|_2^2.$$

Let $H_{\mathbf{n}} = \prod_{j \in J_K} H_{n_j}$ be the normalized Hermite basis of $L^2(\mu_{\leq K})$. Integration by parts against the Gaussian density gives

$$q_{0,K}(\psi) = \sum_{\mathbf{n} \in \mathbb{N}^{n_K}} \left(\sum_j \gamma_j n_j \right) |\langle H_{\mathbf{n}}, \psi \rangle|^2.$$

Consequently, for a set bounded by $q_K(\psi) \leq R$ and any $A > 0$,

$$\sum_{\sum_j \gamma_j n_j > A} |\langle H_{\mathbf{n}}, \psi \rangle|^2 \leq R/A.$$

Only finitely many multi-indices satisfy $\sum_j \gamma_j n_j \leq A$. Projection onto those indices is finite dimensional, while the displayed tail is uniform; hence the form-domain unit ball is relatively compact in L^2 . The retained-mode operator therefore has compact resolvent for every $\lambda \geq 0$. Equation (11) and the self-adjoint free tail then give a self-adjoint, lower-bounded $K^{(K)}$.

For (ii), the reference dynamics is the n_K -mode OU semigroup $P_t^{\leq K} = \bigotimes_j M_t^{(j)}$ with the Mehler kernels

$$M_t^{(j)}(\zeta, \zeta') = (1 - r_j^2)^{-1/2} \exp \left[-\frac{r_j^2(\zeta^2 + \zeta'^2) - 2r_j\zeta\zeta'}{2(1 - r_j^2)} \right], \quad r_j := e^{-t\gamma_j}, \quad (17)$$

with respect to μ_1 in each coordinate. Completing the square gives $\int M_t^{(j)}(\zeta, \zeta') d\mu_1(\zeta') = 1$; the kernel is symmetric and strictly positive. The Gaussian generating function for the normalized Hermite polynomials gives eigenvalue r_j^n in degree n . Hence the product kernel is exactly $\Gamma(e^{-t\gamma})$ on the retained Gaussian space.

For $R > 0$ let $W_R = W \wedge R$. Since $-b_K \leq W_R \leq R$, it is a bounded self-adjoint multiplication operator, and the bounded-perturbation Trotter formula gives

$$e^{-t(H_0^{\leq K} + W_R)} = s \lim_{n \rightarrow \infty} (e^{-(t/n)H_0^{\leq K}} e^{-(t/n)W_R})^n. \quad (18)$$

Inserting the product Mehler kernel between the n multiplication factors turns a bounded matrix element into the OU chain integral with weight $\exp[-(t/n) \sum_{\ell=1}^n W_R(X_{\ell t/n})]$. The finite-dimensional OU process has continuous paths and W_R is continuous, so the exponent is a Riemann sum converging almost surely to $-\int_0^t W_R(X_s) ds$; it is dominated by e^{tb_K} . Dominated convergence proves (12) with W_R . As $R \rightarrow \infty$, $W_R \uparrow W$; monotone form convergence gives the operator limit, and the path weights decrease to $e^{-\int_0^t W(X_s) ds}$ while remaining bounded by e^{tb_K} . A second dominated-convergence step proves (12). Conditioning on the endpoints expresses its kernel as the strictly positive continuous Mehler endpoint density times a bridge expectation in $(0, e^{tb_K}]$. Realize the bridge as an affine continuous function of the endpoints plus an endpoint-independent centered bridge; dominated convergence proves joint continuity.

The strictly positive compact kernel also proves simplicity without an omitted Perron–Frobenius step. Put $T = e^{-t(H_0^{\leq K} + W)}$ and $r = \|T\| > 0$. Compactness and positivity of the self-adjoint operator give a real eigenvector u with $Tu = ru$. Replacing it by $|u|$ is legitimate because

$$r\|u\|_2 = \|Tu\|_2 \leq \|T|u|\|_2 \leq r\||u||_2 = r\|u\|_2;$$

the spectral expansion of the nonnegative operator T shows that equality in the last inequality makes $|u|$ an r -eigenvector. The strictly positive kernel gives $|u| > 0$ a.e.; rename it u .

If the r -eigenspace had dimension at least two, reality of the kernel would give a nonzero real r -eigenvector $h \perp u$. Then both $h_+ = \max(h, 0)$ and $h_- = \max(-h, 0)$ are nonzero (otherwise $\langle u, h \rangle$ could not vanish). Strict positivity of the kernel gives $Th_+ > 0$ and $Th_- > 0$ a.e., and hence

$$r|h| = |Th_+ - Th_-| < Th_+ + Th_- = T|h| \quad \text{a.e.}$$

Pairing with $u > 0$ and using self-adjointness yields the contradiction

$$r\langle u, |h| \rangle < \langle u, T|h| \rangle = \langle Tu, |h| \rangle = r\langle u, |h| \rangle.$$

Thus the retained lowest eigenvalue is simple. The free tail has its unique vacuum, proving the simplicity assertion in (i).

(iii) Tensor with the free factor: $T_K(t) = e^{-t(H_0^{\leq K} + V^{(K)})} \otimes \Gamma(e^{-t\gamma})|_{>K}$, and for the free factor the OU pair formula (5) applies; the product of the two expectations is the joint expectation because the $\leq K$ and $> K$ OU coordinates are independent.

(iv) By the form inequality $H_0^{\leq K} + V^{(K)} \geq H_0^{\leq K} - b_K$ and the min-max principle, $\text{spec}(H_0^{\leq K} + V^{(K)}) \ni \mu_j^{(K)} \geq \nu_j^{\leq K} - b_K$ against the $\leq K$ free eigenvalues, so $\text{Tr} T_K(t) \leq e^{tb_K} Z_0(t)$.

(v) We spell out the cyclic normalization. For bounded W_R and $\delta = t/n$, the trace of the Trotter product in (18) is

$$\int_{(\mathbb{R}^{n_K})^n} \prod_{\ell=0}^{n-1} M_\delta(\zeta_\ell, \zeta_{\ell+1}) e^{-\delta W_R(\zeta_{\ell+1})} \prod_{\ell=0}^{n-1} d\mu_{\leq K}(\zeta_\ell), \quad \zeta_n = \zeta_0. \quad (19)$$

With $W_R = 0$, this integral is $Z_{0, \leq K}(t) = \prod_{j \in J_K} (1 - e^{-t\gamma_j})^{-1}$. Divide (19) by that number. Its covariance can be inverted exactly before taking a limit. For one mode put $\delta = t/n$ and $r = e^{-\delta\gamma_j}$. Expanding the product of the n Mehler kernels against $\prod_e \text{ld}\mu_1(\zeta_{e\ell})$ gives the cyclic precision matrix

$$A_n = (1 - r^2)^{-1} [(1 + r^2)I - r(S + S^*)],$$

where S is the cyclic right shift. For $0 \leq d \leq n$ set

$$C_d := \frac{r^d + r^{n-d}}{1 - r^n}.$$

For $1 \leq d < n$, $(1 + r^2)C_d - rC_{d-1} - rC_{d+1} = 0$, while $(1 + r^2)C_0 - 2rC_1 = 1 - r^2$. Hence the cyclic matrix $(C_{|a-b|})_{a,b}$ is A_n^{-1} . At mesh times $s = a\delta$, $s' = b\delta$ this gives, exactly,

$$\mathbb{E}[\xi_j(s)\xi_j(s')] = \frac{e^{-\gamma_j|s-s'|} + e^{-\gamma_j(t-|s-s'|)}}{1 - e^{-t\gamma_j}} = \frac{\cosh(\gamma_j(t/2 - |s - s'|))}{\sinh(\gamma_j t/2)}.$$

These consistent mesh marginals converge to the continuous periodic Gaussian process. Equivalently, its covariance has the temporal Fourier expansion

$$\mathbb{E}[\xi_j(s)\xi_j(s')] = \frac{2\gamma_j}{t} \sum_{k_0 \in (2\pi/t)\mathbb{Z}} \frac{e^{ik_0(s-s')}}{k_0^2 + \gamma_j^2} = \frac{\cosh(\gamma_j(t/2 - |s - s'|_{\mathbb{T}_t}))}{\sinh(\gamma_j t/2)}. \quad (20)$$

where the first equality is the Fourier series of the preceding continuous function and the second is the Mittag-Leffler sum. Since the field coefficient is $\xi_j/\sqrt{2\gamma_j}$, (20) is exactly the covariance c_{γ_j} in Definition 2.4(ii).

Let $v_K(t)$ be this periodic field's equal-point variance and $d^{(K)} = v_K - c_K$. The two polynomial identities

$$:X^4:_{c_K} = :X^4:_{v_K} + 6d^{(K)}:X^2:_{v_K} + 3(d^{(K)})^2, \quad :X^2:_{c_K} = :X^2:_{v_K} + d^{(K)}$$

show that the limiting exponent in (19) is precisely $\mathcal{U}_t^{(K)}$ as displayed in (9). The limits $n \rightarrow \infty$ and then $R \rightarrow \infty$ also commute with the trace, as follows directly from the diagonal kernels. For every R the positive continuous Feynman–Kac kernel satisfies

$$0 \leq K_R(t; \zeta, \zeta) \leq e^{tb_K} M_t(\zeta, \zeta), \quad \int M_t(\zeta, \zeta) d\mu_{\leq K}(\zeta) = Z_{0, \leq K}(t) < \infty.$$

The trace equals the diagonal integral without a basis interchange: symmetry and the semigroup identity give $K_R(t; \zeta, \zeta) = \int K_R(t/2; \zeta, z)^2 d\mu(z)$; integrating and using Tonelli gives $\int K_R(t; \zeta, \zeta) d\mu(\zeta) = \|e^{-t(H_0 + W_R)/2}\|_{\mathcal{S}_2}^2 = \text{Tre}^{-t(H_0 + W_R)}$. Thus dominated convergence applies first to the Trotter-chain diagonal and then, because $W_R \uparrow W$, to $K_R(t; \zeta, \zeta) \downarrow K(t; \zeta, \zeta)$. This proves the retained-mode trace identity without inferring trace convergence from strong convergence. Tensoring the independent free tail multiplies $Z_{0, \leq K}(t)$ by $Z_{0, > K}(t)$, yielding

$$\text{Tr} T_K(t) = Z_0(t) \mathbb{E}_{\mu_{t,L}^{\text{per}}} [e^{-\mathcal{U}_t^{(K)}}].$$

(vi) It remains to account for the Weyl insertion. In an ordinary retained Schrödinger coordinate the BCH identity gives, with real source coefficients α, β ,

$$(W(\alpha, \beta)\psi)(q) = e^{i\alpha q + i\alpha\beta/2} \psi(q + \beta).$$

Indeed, $[i\alpha q, \beta \partial_q] = -i\alpha\beta$ is central, so $e^{i\alpha q + \beta \partial_q} = e^{i\alpha\beta/2} e^{i\alpha q} e^{\beta \partial_q}$. Therefore the trace kernel joins q_0 to $q_t = q_0 + \beta$. Set $q_s = \eta_s + H_s$ with $\eta_t = \eta_0$, $H_0 = -\beta/2$, and $H_t = +\beta/2$. The Weyl phase becomes

$$e^{i\alpha q_0 + i\alpha\beta/2} = e^{i\alpha\eta_0};$$

thus the CCR half-phase cancels exactly, leaving the sharp-time position source.

For each Fourier mode the unique function with those endpoint values, $\dot{H}(t) = \dot{H}(0)$, and $-\ddot{H} + \gamma(k)^2 H = 0$ is

$$H_{p,k}^{(K)}(s) = \frac{b_\infty(k)}{2} \frac{\sinh(\gamma(k)(s - t/2))}{\sinh(t\gamma(k)/2)}, \text{quad } |k| \leq K. \quad (21)$$

The Gaussian shift is exact already on every Mehler time mesh. At an interior mesh node the two cross coefficients are $-\dot{H}(s_i)$ from the factor on its left and $+\dot{H}(s_i)$ from the factor on its right; they cancel. At the marked node their sum is $\dot{H}(t) - \dot{H}(0) = 0$. Hence no linear term remains, and the constants sum to the Cameron–Martin action. Equivalently, for smooth periodic η ,

$$\Re \int_0^t (\bar{\eta} \dot{H} + \gamma(k)^2 \bar{\eta} H) ds = \Re [\bar{\eta} \dot{H}]_0^t + \Re \int_0^t \bar{\eta} (-\ddot{H} + \gamma(k)^2 H) ds = 0;$$

the mesh identity is the rigorous statement for OU paths.

Spatial Fourier Parseval is $\frac{1}{2} \int_{\mathbb{T}_L} (|\partial_s H|^2 + |\gamma H|^2) dx = \frac{1}{4L} \sum_k (|\dot{H}_k|^2 + \gamma(k)^2 |H_k|^2)$. For the remaining constant of one Fourier mode, integration by parts gives

$$\begin{aligned} \frac{1}{4L} \int_0^t (|\dot{H}|^2 + \gamma(k)^2 |H|^2) ds &= \frac{1}{4L} \Re [\bar{H} \dot{H}]_0^t \\ &= \frac{1}{4L} \left(\frac{\gamma(k) |b_\infty(k)|^2}{4} \coth \frac{t\gamma(k)}{2} + \frac{\gamma(k) |b_\infty(k)|^2}{4} \coth \frac{t\gamma(k)}{2} \right) \\ &= \frac{1}{8L} \gamma(k) \coth \left(\frac{t\gamma(k)}{2} \right) |b_\infty(k)|^2. \end{aligned}$$

Thus

$$S_k(H_p) = \frac{1}{8L} \gamma(k) \coth(t\gamma(k)/2) |b_\infty(k)|^2.$$

Summing the retained modes is the action in the theorem. Completing the Gaussian square therefore contributes $e^{-S^{(K)}}$. The position component of the Weyl operator remains as the sharp-time phase $e^{i\Phi(J_q)}$. Finally, the exact Appell identity

$$:(X + H)^4:_c = :X^4:_c + 4H:X^3:_c + 6H^2:X^2:_c + 4H^3X + H^4$$

(and its quadratic analogue) identifies the translated interaction with $\mathcal{U}_t^{(K)}(\eta + H_p^{(K)})$. The bounded truncation, Trotter-mesh, and $R \rightarrow \infty$ limits use the same domination e^{tb_K} ; the Weyl insertion has norm one. Since the Weyl data have no tail modes, the independent free-tail trace factor is identical in numerator and denominator and cancels. This proves (13) with every normalization displayed. $\square$

Remark 3.2 (finite-mode dictionary). Equations (14)–(15) give the unitary coordinate identification, (16) proves the weighted form realization, (17)–(18) prove FKN, and (19)–(20) fix the trace normalization. Thus every finite-mode step used later is contained in the displayed Gaussian calculation; V1 supplies matching notation, not an additional unwritten theorem.

4 The Hamiltonian K_L (A1-i)

Theorem 4.1 (construction of K_L). *There is a unique self-adjoint, bounded-below operator K_L on $L^2(Q, \mu)$ such that $T_K(t) \rightarrow e^{-tK_L} =: T(t)$ strongly for every $t \geq 0$. It satisfies: (i) $\inf \text{spec } K_L \geq -c_\lambda$ and $K_L \geq \frac{1}{2}H_0 - c_{1/2}$ in the sense of quadratic forms; (ii) for every bounded cylinder f, g ,*

$$\langle f, T(t)g \rangle = \mathbb{E} \left[\overline{f(X_0)} g(X_t) e^{-\mathcal{W}} \right], \quad \mathcal{W} = \int_0^t V_L(X_s) ds, \quad (22)$$

and (22) extends to all bounded measurable f, g on Q ; (iii) $\mathcal{D} \subset D(K_L)$ with $K_L\psi = H_0\psi + V_L\psi$ for $\psi \in \mathcal{D}$ (operator sum; $V_L\psi \in L^2$ since ψ is bounded); (iv) $\mathcal{D} \subset Q(K_L)$ with $q_{K_L}(\psi) = \langle \psi, H_0\psi \rangle + \int V_L|\psi|^2 d\mu$ there (form sum).

Proof. Strong Cauchy property. For $\psi \in \mathcal{D}$ (based on modes $\leq K_0$) and $K, K' \geq K_0$, conditional Cauchy–Schwarz on (12) gives

$$\|(T_K(t) - T_{K'}(t))\psi\|_2^2 = \mathbb{E} \left[\left| \mathbb{E}[\psi(X_t)(e^{-\mathcal{W}_K} - e^{-\mathcal{W}_{K'}}) \mid X_0] \right|^2 \right] \leq \|\psi\|_\infty^2 \|e^{-\mathcal{W}_K} - e^{-\mathcal{W}_{K'}}\|_{L^2}^2,$$

and by $|e^{-\alpha} - e^{-\beta}| \leq |\alpha - \beta|(e^{-\alpha} + e^{-\beta})$, Hölder, Proposition 2.6(ii) and the chaos moment bound,

$$\|e^{-\mathcal{W}_K} - e^{-\mathcal{W}_{K'}}\|_{L^2} \leq \|\mathcal{W}_K - \mathcal{W}_{K'}\|_{L^4} (\|e^{-\mathcal{W}_K}\|_{L^8} + \|e^{-\mathcal{W}_{K'}}\|_{L^8}) \leq Ct \|V^{(K)} - V^{(K')}\|_{L^2(\mu)}, \quad (23)$$

uniformly for $t \leq T$. With (4) and $\|T_K(t)\| \leq e^{tc_\lambda}$ (Corollary 2.7) the family $T_K(t)$ is strongly Cauchy on $\mathcal{D}$, hence on L^2 ; call the limit $T(t)$.

Semigroup, self-adjointness, strong continuity. Each T_K is a self-adjoint contraction semigroup up to e^{tc_λ} ; the limit inherits $T(t)^* = T(t)$, $T(t)T(t') = T(t+t')$ (strong limits on the uniformly bounded family), and $\|T(t)\| \leq e^{tc_\lambda}$. Strong continuity at $t = 0$: for $\psi \in \mathcal{D}$ and any $K \geq K_0$, $\|T(t)\psi - \psi\| \leq \|T(t)\psi - T_K(t)\psi\| + \|T_K(t)\psi - \psi\|$; the first term is bounded, uniformly in $t \leq 1$, by $C\|\psi\|_\infty t \cdot \sup_{K' \geq K} \|V^{(K)} - V^{(K')}\|_2 \leq C'\|\psi\|_\infty \varrho_K$ with $\varrho_K \rightarrow 0$, by

(23) (note the factor $t \leq 1$); the second tends to 0 as $t \downarrow 0$ for fixed K (C_0 -semigroup). Hence $\limsup_{t \downarrow 0} \|T(t)\psi - \psi\| \leq C' \|\psi\|_\infty \varrho_K$ for every K , i.e. $= 0$; density and uniform bounds finish. By the Hille–Yosida theorem for self-adjoint semigroups, $T(t) = e^{-tK_L}$ with K_L self-adjoint, $K_L \geq -c_\lambda$.

(i). The first bound is $\|T(t)\| \leq e^{tc_\lambda}$. For the second we use resolvent antitonicity, which avoids any form-core question (none is available: $\mathcal{D}$ is *not* shown to be a form core, cf. Remark 4.2). Fix $M > c_{1/2} \vee c_\lambda$ and recall the Legendre identity: for a self-adjoint $S \geq \varepsilon > 0$ with form q_S (set $q_S(\psi) := +\infty$ off $Q(S)$),

$$q_S(\psi) = \sup_{\phi \in \mathcal{H}} [2\operatorname{Re}\langle \psi, \phi \rangle - \langle \phi, S^{-1}\phi \rangle], \quad \langle \phi, S^{-1}\phi \rangle = \sup_{\psi \in \mathcal{H}} [2\operatorname{Re}\langle \psi, \phi \rangle - q_S(\psi)], \quad (24)$$

both by the spectral theorem and the scalar identity $\sup_c [2\operatorname{Re}(\bar{\psi}c) - |c|^2/\lambda] = \lambda|\psi|^2$ ($\lambda > 0$), applied componentwise. Consequently, for self-adjoint $S_1, S_2 \geq \varepsilon$: $q_{S_1} \geq q_{S_2}$ pointwise on $\mathcal{H}$ (with the $+\infty$ convention, which in particular forces $Q(S_1) \subseteq Q(S_2)$) *if and only if* $S_1^{-1} \leq S_2^{-1}$ — each direction is one application of (24) under the supremum. Now $K^{(K)} + M \geq \frac{1}{2}H_0 + (M - c_{1/2})$ as forms (Corollary 2.7), so $(K^{(K)} + M)^{-1} \leq (\frac{1}{2}H_0 + M - c_{1/2})^{-1}$ for every K . Strong semigroup convergence and $\|T_K(t)\| \leq e^{tc_\lambda}$ give, by dominated convergence in the Laplace representation $(K^{(K)} + M)^{-1} = \int_0^\infty e^{-Mt} T_K(t) dt$, strong resolvent convergence $(K^{(K)} + M)^{-1} \rightarrow (K_L + M)^{-1}$. Hence $(K_L + M)^{-1} \leq (\frac{1}{2}H_0 + M - c_{1/2})^{-1}$, and the “if” direction of the equivalence (take the supremum over ϕ in the first Legendre identity) returns $K_L \geq \frac{1}{2}H_0 - c_{1/2}$ as quadratic forms, including $Q(K_L) \subseteq Q(H_0)$.

(ii). Fix bounded cylinder f, g based on modes $\leq K_0$. By Theorem 3.1(iii), $\langle f, T_K(t)g \rangle = \mathbb{E}[\bar{f}(X_0)g(X_t)e^{-\mathcal{W}_K}]$ for $K \geq K_0$; the left side converges to $\langle f, T(t)g \rangle$, the right side to (22) by (23) ($\mathbb{E}[\|f\bar{g}\|e^{-\mathcal{W}_K} - e^{-\mathcal{W}}] \leq \|f\|_\infty \|g\|_\infty \|e^{-\mathcal{W}_K} - e^{-\mathcal{W}}\|_{L^1}$). Extension to bounded measurable f, g : both sides are continuous under bounded pointwise-a.e. convergence (dominated on the right by $\|f\|_\infty \|g\|_\infty e^{-\mathcal{W}} \in L^1$; strong on the left), and bounded cylinder functions are dense in that sense (martingale convergence for the mode filtration).

(iii). Let $\psi \in \mathcal{D}$. Write, using (22) and the free OU semigroup $P_s = \Gamma(e^{-s\gamma})$,

$$\frac{(1 - T(t))\psi}{t} = \frac{(1 - P_t)\psi}{t} - \frac{1}{t} \mathbb{E}[\psi(X_t)(e^{-\mathcal{W}} - 1) \mid X_0 = \cdot].$$

The first term converges in L^2 to $H_0\psi$ ($\psi \in \mathcal{D} \subset D(H_0)$). For the second, expand $e^{-\mathcal{W}} - 1 = -\mathcal{W} + \rho$, $|\rho| \leq \frac{1}{2}\mathcal{W}^2 e^{\mathcal{W}_-}$ with $\mathcal{W}_- := \max(-\mathcal{W}, 0)$:

$$\frac{1}{t} \mathbb{E}[\psi(X_t)\mathcal{W} \mid X_0] = \frac{1}{t} \int_0^t P_s(V_L \cdot P_{t-s}\psi) ds \xrightarrow[t \downarrow 0]{L^2} V_L \psi,$$

because $\|P_s(V_L P_{t-s}\psi) - V_L \psi\|_2 \leq \|V_L(P_{t-s}\psi - \psi)\|_2 + \|(P_s - 1)(V_L \psi)\|_2 \leq \|V_L\|_{L^4} \|P_{t-s}\psi - \psi\|_{L^4} + o(1) \rightarrow 0$ uniformly in $s \leq t$ (for $\psi = F(\xi_1, \dots, \xi_n) \in \mathcal{D}$ the Mehler representation gives the pointwise bound $|P_u \psi - \psi| \leq \operatorname{Lip}(F) \sum_{j \leq n} [(1 - e^{-u\gamma_j})|\xi_j| + (1 - e^{-2u\gamma_j})^{1/2} \mathbb{E}|\zeta_j|]$ with ζ_j standard Gaussian, whence $\sup_{u \leq t} \|P_u \psi - \psi\|_{L^4(\mu)} \leq C_n \operatorname{Lip}(F) \sum_{j \leq n} (1 - e^{-t\gamma_j})^{1/2} \rightarrow 0$ as $t \downarrow 0$ — an L^4 , not a uniform-on- Q , statement, which is all that is used; and $V_L \psi \in L^2$ with P_s strongly continuous). The remainder: by Cauchy–Schwarz and Proposition 2.6(ii), $t^{-1} \|\mathbb{E}[\psi(X_t)\rho \mid X_0]\|_{L^2} \leq t^{-1} \|\psi\|_\infty \|\mathcal{W}^2 e^{\mathcal{W}_-}\|_{L^2} \leq t^{-1} \|\psi\|_\infty \|\mathcal{W}\|_{L^2}^2 \|e^{\mathcal{W}_-}\|_{L^4} \leq C t^{-1} \cdot t^2 = Ct \rightarrow 0$ (here $\|\mathcal{W}\|_{L^2} \leq 49t \|V_L\|_2$ by stationarity and the chaos bound, and $e^{\mathcal{W}_-} \leq 1 + e^{-\mathcal{W}}$ with the uniform L^4 bound). Hence $t^{-1}(1 - T(t))\psi \rightarrow H_0\psi + V_L \psi$ in L^2 , which is precisely $\psi \in D(K_L)$ with $K_L \psi = H_0\psi + V_L \psi$.

(iv). For $\psi \in \mathcal{D}$, spectral calculus gives $t^{-1} \langle \psi, (1 - T(t))\psi \rangle \uparrow q_{K_L}(\psi) \in [-c_\lambda \|\psi\|^2, \infty]$ as $t \downarrow 0$ (monotone because $t \mapsto (1 - e^{-tx})/t$ is decreasing for every $x \in \mathbb{R}$, its t -derivative having numerator $e^{-tx}(1 + tx) - 1 \leq 0$; the lower endpoint is $\inf \operatorname{spec} K_L \geq -c_\lambda$, not 0, the

ground energy μ_0 being in general negative); by (iii) the limit equals $\langle \psi, (H_0 + V_L)\psi \rangle = q_0(\psi) + \int V_L |\psi|^2 d\mu$, finite. (Part (i) uses no such $t \downarrow 0$ interchange: it is proved by resolvent antitonicity through the Legendre identity and the Laplace representation. The interchange above is used only here.)

Uniqueness is immediate: the semigroup determines its generator. $\square$

Remark 4.2 (on the “form sum” clause and the literature object). Theorem 4.1(iii)–(iv) realize the A1(i) clause: K_L is a self-adjoint, bounded-below extension of the operator sum $H_0 + V_L$ from the cylinder core, whose quadratic form is the form sum there. Two scope notes, recorded for the ledger. (1) Whether $(H_0 + V_L)|_{\mathcal{D}}$ is *essentially* self-adjoint (Segal’s theorem in the $\mathbb{T}_L$ setting; cf. Simon’s book, §V, in [refs/](#)) is not proved here and *not consumed anywhere in Part II*: every downstream use of K_L (A2(b), A6–A8) goes through the semigroup $T(t)$, its traces, and its ground state, i.e. through this construction. (2) Consequently $\omega_{L,\lambda}^{\text{ct}}$ (Theorem 7.1) is *defined* by this K_L ; this is the same normalization the scoping note fixed when it declared A1 the definition of the gate’s right-hand side.

5 Trace class and trace-norm cylinder approximation (A1-ii)

Lemma 5.1 (uniform hypercontractive smoothing). *Let $s_4 := \log 3/m$. Then, uniformly in K (and in the bounded-truncation parameter R where it appears in the proof),*

$$\|T_K(s)\|_{L^2 \rightarrow L^4} \leq M_0 e^{c\lambda(s-s_4)} \quad (s \geq s_4),$$

with $M_0 = M_0(\lambda, a, b, m, L) < \infty$; the same bound holds for $T(s)$.

Proof. By duality and self-adjointness it suffices to bound $\|T_K(s_4)\|_{L^{4/3} \rightarrow L^2}$ ($s > s_4$ then composes with $\|T_K(s - s_4)\|_{2 \rightarrow 2} \leq e^{c\lambda(s-s_4)}$, applied first). Let $W := V^{(K)}$, $W_R := W \wedge R$, and let $K_R^{(K)}$ be the form sum with W_R ; by monotone form convergence $e^{-sK_R^{(K)}} \rightarrow e^{-sK^{(K)}}$ strongly as $R \rightarrow \infty$ (model-sheet §3 tool), so by Fatou (pointwise a.e. along a subsequence) it suffices to bound $\|e^{-s_4 K_R^{(K)}}\|_{4/3 \rightarrow 2}$ uniformly in K, R . By the elementary Trotter product formula (self-adjoint semibounded + bounded; the model-sheet (3.16) instance) and Fatou again, it suffices to bound the products $(e^{-\tau W_R} e^{-\tau H_0})^n$ uniformly, $\tau := s_4/n$.

Set $p_j := 1 + \frac{1}{3}e^{j\tau m}$, $j = 0, \dots, n$, so $p_0 = 4/3$ and $p_n = 2$. Each free factor is bounded by Import 1.1(d): $\|e^{-\tau H_0}\|_{L^{p_j} \rightarrow L^{P_j}} \leq 1$ with $P_j := 1 + (p_j - 1)e^{2\tau m}$ (since $\|e^{-\tau \gamma}\| \leq e^{-\tau m}$). Each multiplication costs, by Hölder with $\frac{1}{p_{j+1}} = \frac{1}{r_j} + \frac{1}{P_j}$,

$$\begin{aligned} \|e^{-\tau W_R} u\|_{p_{j+1}} &\leq \|e^{-\tau W_R}\|_{L^{r_j}} \|u\|_{P_j}, \\ \|e^{-\tau W_R}\|_{L^{r_j}} &= \mathbb{E}[e^{-r_j \tau W_R}]^{1/r_j} \leq (1 + \mathbb{E}[e^{-r_j \tau W}])^{1/r_j}, \end{aligned}$$

using $e^{-\tau W_R} \leq 1 + e^{-\tau W}$ pointwise. The exponent budget: $P_j - p_{j+1} = \frac{1}{3}e^{(j+1)\tau m}(e^{\tau m} - 1) > 0$, and, using $p_{j+1} \geq \frac{4}{3}$, $P_j \geq \frac{4}{3}$ and $e^{n\tau m} = 3$,

$$\frac{1}{r_j} = \frac{P_j - p_{j+1}}{p_{j+1}P_j} \leq \frac{9}{16} \cdot \frac{e^{(j+1)\tau m}(e^{\tau m} - 1)}{3}, \quad \sum_{j=0}^{n-1} \frac{1}{r_j} \leq \frac{3}{16} e^{\tau m} (e^{n\tau m} - 1) = \frac{3}{8} e^{\tau m} \leq 1$$

for $n \geq 2$ (then $e^{\tau m} \leq \sqrt{3}$). For the Hölder *times*, the crude bound $p_{j+1}P_j \leq 4$ would be *false* at the top rungs ($p_{n-1} = 1 + e^{\tau m} > 2$); instead use $p_{j+1} \leq 2$, $P_j \leq 1 + \frac{1}{3}e^{(j+2)\tau m}$ and $\tau m \leq e^{\tau m} - 1$:

$$\begin{aligned} r_j \tau m &= \tau m \frac{p_{j+1}P_j}{P_j - p_{j+1}} = \frac{3\tau m p_{j+1}P_j}{e^{(j+1)\tau m}(e^{\tau m} - 1)} \\ &\leq 3p_{j+1} \left(e^{-(j+1)\tau m} + \frac{1}{3}e^{\tau m} \right) \leq 6 + 2e^{\tau m} \leq 6 + 2\sqrt{3} < 12, \end{aligned}$$

so $r_j\tau \leq 12/m$ for all j ; hence each expectation is bounded by

$$D := 1 + \sup_K \sup_{0 < \sigma \leq 12/m} \mathbb{E}[e^{-\sigma V^{(K)}}] < \infty$$

(Proposition 2.6(i) at $\sigma_0 = 12/m$; the supremum over $0 < \sigma \leq \sigma_0$ follows from that single bound because $e^{-\sigma V} \leq 1 + e^{-\sigma_0 V}$ pointwise for $0 < \sigma \leq \sigma_0$). Multiplying the factors, $\|(e^{-\tau W_R} e^{-\tau H_0})^n\|_{4/3 \rightarrow 2} \leq D \Sigma^{1/r_j} \leq D \vee 1 =: M_0$, uniformly in n, K, R . (The tensor free factor $> K$ rides along inside H_0 ; no separate argument is needed since W is $\leq K$ -cylindrical and Import 1.1(d) is applied to the full $\Gamma(e^{-\tau \gamma})$.) The bound for $T(s)$ follows by strong convergence and Fatou. $\square$

Theorem 5.2 (A1-ii). *For every $t > 0$: e^{-tK_L} is trace class; $\|T_K(t) - T(t)\|_{\mathcal{J}_1} \rightarrow 0$; $\text{Tr } T_K(t) \rightarrow \text{Tr } T(t)$; the spectrum of K_L is discrete, $\mu_0 \leq \mu_1 \leq \dots \rightarrow \infty$ with $\mu_j \geq \frac{1}{2}\nu_j - c_{1/2}$, and each $\mu_j^{(K)} \rightarrow \mu_j$ as $K \rightarrow \infty$. Moreover, for every bounded operator B , $\text{Tr}(T_K(t)B) \rightarrow \text{Tr}(T(t)B)$.*

Proof. Step 1: norm convergence at $t_2 := 2s_4$.

Step 1a: both Duhamel vectors lie in both form domains. This is the one point where the Duhamel identity needs an argument: differentiating $s \mapsto \langle T_K(s)\phi, T_{K'}(t_2 - s)\psi \rangle$ produces $\langle T_K(s)\phi, K^{(K')}T_{K'}(t_2 - s)\psi \rangle - \langle K^{(K)}T_K(s)\phi, T_{K'}(t_2 - s)\psi \rangle$, and rewriting this as $q_{K^{(K')}} - q_{K^{(K)}}$ evaluated on the pair, so that the H_0 parts cancel and only $\int \Delta V \bar{u}v d\mu$ survives, requires *each* vector to lie in *both* form domains. Membership in $Q(H_0)$ is automatic: $D(K^{(K)}) \subseteq Q(K^{(K)}) \subseteq Q(H_0)$, the second inclusion because $K^{(K)} \geq \frac{1}{2}H_0 - c_{1/2}$ as quadratic forms *uniformly in K* — this is the finite-mode statement of Corollary 2.7, from which Theorem 4.1(i) then derives the same for the limit K_L ; it is the finite-mode form that is needed here, but $Q(H_0) \not\subseteq Q(V^{(K')})$ in general, so it must be supplied. Take $\phi, \psi \in \mathcal{D}$, in particular bounded. The Feynman–Kac formula (12) gives $|(T_K(s)\phi)(x)| \leq \|\phi\|_\infty \mathbb{E}_x[e^{-W_K}]$, and $x \mapsto \mathbb{E}_x[\cdot]$ is a conditional expectation on the stationary OU ensemble, so by conditional Jensen and Proposition 2.6(ii),

$$\|T_K(s)\phi\|_{L^4(\mu)} \leq \|\phi\|_\infty \|e^{-W_K}\|_{L^4} \leq C \|\phi\|_\infty \quad \text{uniformly in } K \text{ and } s \in [0, t_2],$$

and likewise for $T_{K'}(t_2 - s)\psi$. Since $V^{(K)}, V^{(K')} \in L^2(\mu)$, Cauchy–Schwarz gives, for every $u \in L^4(\mu)$,

$$\int |V^{(\cdot)}| |u|^2 d\mu \leq \|V^{(\cdot)}\|_{L^2} \|u\|_{L^4}^2 < \infty,$$

i.e. $L^4(\mu) \subseteq Q(V^{(K)}) \cap Q(V^{(K')})$. Both vectors therefore lie in $Q(H_0) \cap Q(V^{(K)}) \cap Q(V^{(K')})$, the H_0 parts cancel, and the derivative equals $\int \Delta V \overline{T_K(s)\phi} T_{K'}(t_2 - s)\psi d\mu$, integrable over $s \in [0, t_2]$ by the bounds below.

Step 1b. Integrating that derivative over $[0, t_2]$ gives, for $\phi, \psi \in \mathcal{D}$ and hence by density for all $\phi, \psi \in L^2$,

$$|\langle \phi, (T_K(t_2) - T_{K'}(t_2))\psi \rangle| \leq \int_0^{t_2} |\langle T_K(s)\phi, \Delta V T_{K'}(t_2 - s)\psi \rangle| ds, \quad \Delta V := V^{(K')} - V^{(K)}.$$

For $s \leq t_2/2$: Hölder $\frac{1}{4} + \frac{1}{2} + \frac{1}{4}$ with $\|T_K(s)\phi\|_2 \leq e^{sc_\lambda} \|\phi\|_2$ and $\|T_{K'}(t_2 - s)\psi\|_4 \leq M_0 e^{c_\lambda t_2} \|\psi\|_2$ (Lemma 5.1, $t_2 - s \geq s_4$); for $s \geq t_2/2$ symmetrically. With $\|\Delta V\|_{L^4} \leq 9\|\Delta V\|_{L^2}$ (chaos bound),

$$\|T_K(t_2) - T_{K'}(t_2)\|_{2 \rightarrow 2} \leq 18 t_2 M_0 e^{2c_\lambda t_2} \|V^{(K)} - V^{(K')}\|_{L^2(\mu)} \xrightarrow{K, K' \rightarrow \infty} 0. \quad (25)$$

Hence $\|T_K(t_2) - T(t_2)\| \rightarrow 0$.

Step 2: eigenvalues. Each $T_K(t_2)$ is compact positive (Theorem 3.1(iv)); by (25) and Weyl's inequality for singular values, $T(t_2)$ is compact and each ordered eigenvalue converges: $e^{-t_2\mu_j^{(K)}} \rightarrow e^{-t_2\mu_j}$, where $e^{-t_2\mu_j}$ enumerates $\text{spec } T(t_2)$; hence $\mu_j^{(K)} \rightarrow \mu_j$ and, by Corollary 2.7 and min-max, $\mu_j^{(K)} \geq \frac{1}{2}\nu_j - c_{1/2}$, so also $\mu_j \geq \frac{1}{2}\nu_j - c_{1/2}$. Since $T(t) = T(t_2)^{t/t_2}$ by the spectral theorem (both defined from K_L), the μ_j are t -independent and exhaust $\text{spec } K_L$ (discreteness).

Step 3: traces for all $t > 0$. By dominated convergence on j -sums ($e^{-t\mu_j^{(K)}} \leq e^{tc_{1/2}}e^{-t\nu_j/2}$, summable to $e^{tc_{1/2}}Z_0(t/2)$),

$$\text{Tr } T_K(t) = \sum_j e^{-t\mu_j^{(K)}} \rightarrow \sum_j e^{-t\mu_j} = \text{Tr } T(t) < \infty.$$

Step 4: $\mathcal{J}_1$ -convergence. With $s := t/2$,

$$\|T_K(s) - T(s)\|_{\mathcal{J}_2}^2 = \text{Tr } T_K(2s) - 2 \text{Tr}(T_K(s)T(s)) + \text{Tr } T(2s).$$

The first term converges to $\text{Tr } T(2s)$ (Step 3). For the mixed term, expand in the eigenbasis (ϕ_n) of $T(s)$ with eigenvalues σ_n ($\sum \sigma_n < \infty$):

$$\text{Tr}(T_K(s)T(s)) = \sum_n \sigma_n \langle \phi_n, T_K(s)\phi_n \rangle \rightarrow \sum_n \sigma_n \langle \phi_n, T(s)\phi_n \rangle = \text{Tr } T(2s)$$

by dominated convergence ($|\langle \phi_n, T_K(s)\phi_n \rangle| \leq e^{sc_\lambda}$, strong convergence pointwise in n). Hence $\|T_K(s) - T(s)\|_{\mathcal{J}_2} \rightarrow 0$, and by Hölder $\|T_K(t) - T(t)\|_{\mathcal{J}_1} \leq (\|T_K(s)\|_{\mathcal{J}_2} + \|T(s)\|_{\mathcal{J}_2})\|T_K(s) - T(s)\|_{\mathcal{J}_2} \rightarrow 0$ (the $\mathcal{J}_2$ norms converge, hence are bounded). Finally $|\text{Tr}((T_K(t) - T(t))B)| \leq \|T_K(t) - T(t)\|_{\mathcal{J}_1}\|B\| \rightarrow 0$. $\square$

6 The periodic Feynman–Kac clause (A1-iv) and the closure of A2(b)

6.1 Euclidean-side limits

Lemma 6.1 (shift data and shifted interactions, $K \rightarrow \infty$). *Fix a coarse symbol $\xi = \varepsilon_M q + ip$ (coarse scale M , refined by the D4 map as in V1/V2) and $t \in [t_0, T]$. Let H_p be the continuum interpolant field (V2, Definition 6.2 with multiplier P_∞ , dispersion γ), $H_p^{(K)}$ its mode truncation ($|k| \leq K$), $S^{(K)}, S_\infty$ the corresponding actions, and $J_q^{(K)}, J_q$ the sharp-time position sources. Then, with $\kappa_0 := \eta/2 > 0$: (i) $0 \leq S_\infty(H_p) - S^{(K)}(H_p) \leq C_\xi (K/m)^{-\eta}$; (ii) $\|\Phi(J_q) - \Phi(J_q^{(K)})\|_{L^2(\mu^{\text{per}})} \leq C_\xi (K/m)^{-\kappa_0}$, so $\|e^{i\Phi(J_q)} - e^{i\Phi(J_q^{(K)})}\|_{L^2} \rightarrow 0$; (iii)*

$$\|\mathcal{U}_t^{(K)}(\eta + H_p^{(K)}) - \mathcal{U}_t(\eta + H_p)\|_{L^2(\mu^{\text{per}})} \leq C_\xi (1 + \log(1 + K/m))^2 (K/m)^{-1/2};$$

(iv) $\sup_K \|e^{-\mathcal{U}_t^{(K)}(\eta + H_p^{(K)})}\|_{L^p} < \infty$ for every $p < \infty$, uniformly over coarse data with $\Sigma_M(p) \leq R$, $\bar{P}_M \leq R'$; and, by (iii) together with the final clause of Lemma 2.5, the untruncated member obeys the same bound, $\|e^{-\mathcal{U}_t(\eta + H_p)}\|_{L^p} < \infty$.

Proof. (i) The action tail is $\frac{1}{4} \sum_{|k| > K} \gamma(k) \coth(\frac{t\gamma}{2}) |\hat{h}(k)|^2$ with the D4 envelope $|\hat{h}(k)|^2 \leq C(1 + \varepsilon_M |k|)^{-2-\eta}$ (V2, §6); since $\coth \leq \coth(t_0 m/2)$ and $\gamma(k) \leq m + |k|$, the tail is bounded

by $C \sum_{|k|>K} |k|^{-1-\eta} \leq C'(K/m)^{-\eta}$. (ii) $\Phi(J_q - J_q^{(K)})$ is Gaussian with variance given by the thermal sharp-time covariance paired against the source tail,

$$\text{Var}[\Phi(J_q - J_q^{(K)})] \leq \coth\left(\frac{t_0 m}{2}\right) \sum_{|k|>K} \frac{|\hat{q}_{\text{tail}}(k)|^2}{2\gamma(k)} \rightarrow 0$$

with the q -envelope, at rate $(K/m)^{-\eta}$; then $|e^{ia} - e^{ib}| \leq |a - b|$. (iii) Expand both shifted interactions by the Wick binomial (V2, Prop. 6.5's display). The chaos- j terms differ through (a) the plain kernel tails — the $j = 4, 2$ parts of Proposition 2.6(iii)'s rate — and (b) the deterministic coefficients $(H^{(K)})^{4-j}$ vs. $(H_p)^{4-j}$. For (b): both fields share the dispersion γ and multiplier family, so $\sup_{s,x} |H_p^{(K)}| \leq \Sigma_M(p)$ (V2, Lemma 6.3(i), whose count is cutoff-monotone) and the mode- ℓ^2 difference of the coefficient fields is a pure tail, $\sup_s \frac{1}{2L} \sum_{|k|>K} |a_k^{(\infty)}(s)|^2 \leq C \bar{P}_M^2(\varepsilon_M K)^{-1-\eta}$ (envelope tail as in V2, Lemma 6.3(iii), with no filter-tail or dispersion terms); the self-convolution telescopes and the normalized power-symbol bounds B_j (V2, Lemma 6.4(i), which is cutoff-uniform) then bound the L^2 differences by $C \Sigma_M(p)^3 \bar{P}_M(K/m)^{-(1+\eta)/2}$ plus the part (a) rate; the slower rate $(K/m)^{-1/2} \log^2$ dominates. (iv) The pointwise lower bound of V2, Lemma 7.1 is invariant under deterministic shifts (a deterministic shift leaves the Wick constant unchanged) and depends on K only through the truncated *vacuum* constant c_K of (2) — the constant that Wick-orders $\mathcal{U}_t^{(K)}$, not the thermal variance — with $c_K \leq C_c \ell_K$, which is (N1); (iii) and the triangle inequality supply (N2) after absorbing one logarithm ($\ell_K \leq C_\zeta(K/m)^\zeta$), i.e. with $\kappa = \frac{1}{2} - \zeta$; and (N0) holds as in Proposition 2.6, $\mathcal{K}$ being again a full tail of Γ_∞ . Lemma 2.5 then applies. $\square$

6.2 The clause

Theorem 6.2 (A1-iv). *For every $t \geq t_0$, with t_0 subject to the standing convention $(\star)$ of §1 (all statements uniform on $t \in [t_0, T]$): (a) (partition identity)*

$$Z_\infty(t) := \text{Tr } e^{-tK_L} = Z_0(t) \mathbb{E}_{\mu_{t,L}^{\text{per}}} [e^{-\mathcal{U}_t}] \in (0, \infty);$$

(b) (cylinder compatibility) $\text{Tr}[e^{-tK^{(K)}} - e^{-tK_L}] \rightarrow 0$ and $\text{Tr}(e^{-tK^{(K)}} B) \rightarrow \text{Tr}(e^{-tK_L} B)$ for every bounded B ; (c) (Weyl insertions, mean-shift form) for every coarse symbol $\xi = \varepsilon_M q + ip$ and its D4 refinement $R_M^\infty \xi$,

$$\frac{\text{Tr}(e^{-tK_L} W_{\text{ct}}(R_M^\infty \xi))}{\text{Tr } e^{-tK_L}} = e^{-S_\infty(H_p)} \frac{\mathbb{E}_{\mu_{t,L}^{\text{per}}} [e^{i\Phi(J_q)} e^{-\mathcal{U}_t(\eta+H_p)}]}{\mathbb{E}_{\mu_{t,L}^{\text{per}}} [e^{-\mathcal{U}_t}]} . \quad (26)$$

Together with $e^{-\mathcal{U}_t}, e^{-\mathcal{U}_t(\eta+H_p)} \in L^p(\mu^{\text{per}})$ for all $p < \infty$ (Proposition 2.6(iii), Lemma 6.1(iv)), this is the full (A1-iv) package of V1, Remark 8.2.

Proof. (b) is Theorem 5.2.

(a). By Theorem 3.1(v), $\text{Tr } T_K(t) = Z_0(t) \mathbb{E}[e^{-\mathcal{U}_t^{(K)}}]$. The left side converges by Theorem 5.2; the right side: by Proposition 2.6(iii) and $|e^{-\alpha} - e^{-\beta}| \leq |\alpha - \beta|(e^{-\alpha} + e^{-\beta})$ with Hölder, $\mathbb{E}[e^{-\mathcal{U}_t^{(K)}}] \rightarrow \mathbb{E}[e^{-\mathcal{U}_t}]$. Positivity: $e^{-\mathcal{U}_t} > 0$ a.s. and $\mathbb{E}[e^{-\mathcal{U}_t}] \geq e^{-\mathbb{E}[\mathcal{U}_t]} > 0$ by Jensen, the ensemble-Wick chaoses being centered, so that $\mathbb{E}[\mathcal{U}_t] = 2Lt \lambda(3d_\infty(t)^2 + a d_\infty(t) + b)$, finite because $0 \leq d_\infty(t) \leq \bar{d} < \infty$ (V2, Lemma 2.1).

(c). Step 1 (finite-mode symbols): if $R_M^\infty \xi$ is replaced by its mode truncation ξ_K (supported in $|k| \leq K$), then Theorem 3.1(vi) gives (26) for the pair (T_K, ξ_K) . Step 2 ($K \rightarrow \infty$ on the Euclidean side): by Lemma 6.1, the numerator converges,

$$\mathbb{E}[e^{i\Phi(J_q^{(K)})} e^{-\mathcal{U}_t^{(K)}(\eta+H_p^{(K)})}] \rightarrow \mathbb{E}[e^{i\Phi(J_q)} e^{-\mathcal{U}_t(\eta+H_p)}]$$

(split the difference into the phase part, bounded by $\|\Phi(J_q - J_q^{(K)})\|_{L^2} \sup_K \|e^{-\mathcal{U}^{(K)}(\cdot)}\|_{L^2}$, and the interaction part, treated as in (a) with Lemma 6.1(iii)–(iv)); the prefactor converges by (i); the denominator as in (a). Step 3 ($K \rightarrow \infty$ on the operator side): $\text{Tr}(T_K(t)W_{\text{ct}}(\xi_K)) - \text{Tr}(T(t)W_{\text{ct}}(R_M^\infty \xi))$ splits into $\text{Tr}((T_K(t) - T(t))W_{\text{ct}}(\xi_K))$, which vanishes by (b) and $\|W\| = 1$, and $\text{Tr}(T(t)(W_{\text{ct}}(\xi_K) - W_{\text{ct}}(R_M^\infty \xi)))$. For the latter: $\xi_K \rightarrow R_M^\infty \xi$ in the one-particle norm (envelope tails), and Weyl operators are strongly continuous along norm-convergent symbols: on the total set of Weyl vectors $W(\zeta)\Omega_0$, $\|(W(\eta_n) - W(\eta))W(\zeta)\Omega_0\|^2 = 2 - 2\text{Re}[e^{i\theta_n}\langle W(\zeta)\Omega_0, W(\eta_n - \eta + \zeta)\Omega_0 \rangle] \rightarrow 0$ by the explicit Fock formula $\langle W(\alpha)\Omega_0, W(\beta)\Omega_0 \rangle = e^{i\sigma(\alpha,\beta)/2} e^{-\|\alpha-\beta\|^2/4}$ and continuity of the symplectic form; unitarity and density upgrade this to strong convergence. Then, expanding $T(t)$ in its eigenbasis, $|\text{Tr}(T(t)(W(\xi_K) - W(R_M^\infty \xi)))| \leq \sum_n e^{-t\mu_n} \|(W(\xi_K) - W(R_M^\infty \xi))\phi_n\| \rightarrow 0$ by dominated convergence. Combining the three steps proves (26). $\square$

Corollary 6.3 (A2(b) closed). *Statement A2(b) of the scoping note, in the exact final form of V1, Definition 8.1, holds for every $t \geq t_0$ with t_0 subject to the standing convention $(\star)$ of §1. (V1’s Definition 8.1 is quantified “for $t > 0$ ”; that range is narrowed here to $t \geq t_0$, since the only support is Theorem 6.2(c), whose proof runs through Lemma 6.1 and Proposition 2.6(iii), both stated on $[t_0, T]$. Nothing downstream is lost: the gate consumes A2(b) only at $t \geq t_* = \max(1, 1/L)$, V4, Theorem 6.2.) The amendment clause (A1-iv) of V1, Remark 8.2 is discharged; no import from Høegh-Krohn (1974) or Figari–Høegh-Krohn–Nappi (1975) is consumed (they remain historical anchors).*

Proof. (26) is V1’s display (Definition 8.1), with S_∞ , H_p , J_q , $\mathcal{U}$ as there; the $e^{-\mathcal{U}} \in L^1$ package required by V1’s chain is Proposition 2.6(iii) and Lemma 6.1(iv). $\square$

7 Gap, ground state, and the state on the Weyl algebra (A1-iii)

Theorem 7.1 (A1-iii). *For every $t > 0$, e^{-tK_L} is positivity improving on $L^2(Q, \mu)$. Consequently the lowest eigenvalue μ_0 of K_L is simple, its normalized eigenvector can be chosen with $\Omega_L > 0$ μ -a.e., and*

$$\gamma_L := \mu_1 - \mu_0 > 0.$$

The state $\omega_{L,\lambda}^{\text{ct}} := \langle \Omega_L, \cdot \Omega_L \rangle$ is well defined on the Weyl algebra $\mathfrak{W}(\mathbb{T}_L)$ generated by $\{W_{\text{ct}}(\zeta)\}$, ζ in the one-particle space $h_{\text{ct}} = H^{-1/2}(\mathbb{T}_L) \oplus iH^{1/2}(\mathbb{T}_L)$ of V1, §6 — on which, and only on which, each $W_{\text{ct}}(\zeta)$ is unitary; $R_M^\infty \xi \in h_{\text{ct}}$ for every coarse ξ , by the D4 envelope (V1, §6).

Proof. Improving. Let $f, g \in L^2$, $f, g \geq 0$, both nonzero; by monotone approximation ($f \wedge n \uparrow f$, positivity preservation of $T(t)$ — itself inherited from Theorem 3.1(ii)’s strictly positive kernels through strong limits) it suffices to take f, g bounded. By (22),

$$\langle f, T(t)g \rangle = \mathbb{E}[f(X_0)g(X_t)e^{-\mathcal{W}}], \quad \mathcal{W} = \int_0^t V_L(X_s)ds.$$

The integrand is nonnegative; $e^{-\mathcal{W}} > 0$ a.s. because $\mathbb{E}|\mathcal{W}| \leq t\|V_L\|_{L^1(\mu)} < \infty$; and with $A := \{f > 0\}$, $B := \{g > 0\}$ (both of positive μ -measure),

$$\mathbb{P}[X_0 \in A, X_t \in B] = \langle \mathbf{1}_A, \Gamma(e^{-t\gamma}) \mathbf{1}_B \rangle_{L^2(\mu)} > 0$$

by (5) and Import 1.1(c) ($\|e^{-t\gamma}\| = e^{-tm} < 1$). On $\{X_0 \in A, X_t \in B\}$ the integrand is strictly positive a.s., so $\langle f, T(t)g \rangle > 0$.

Simplicity and positivity of the ground vector. $T(t)$ is compact (Theorem 5.2), self-adjoint, positive, positivity improving, and commutes with complex conjugation (real FK integrand). Put $r = \|T(t)\| > 0$. Compactness gives an r -eigenvector u ; since

$$r\|u\|_2 = \|T(t)u\|_2 \leq \|T(t)|u|\|_2 \leq r\||u|\|_2,$$

all inequalities are equalities. The spectral expansion of the nonnegative self-adjoint operator $T(t)$ then gives $T(t)|u| = r|u|$, and positivity improvement gives $|u| > 0$ a.e.; take $u = |u|$.

If the r -eigenspace were not one dimensional, commutation with conjugation would provide a nonzero real eigenvector $h \perp u$. Both h_+ and h_- are nonzero, since $u > 0$ and $\langle u, h \rangle = 0$. Positivity improvement gives $T(t)h_+ > 0$ and $T(t)h_- > 0$ a.e., so

$$r|h| = |T(t)h_+ - T(t)h_-| < T(t)h_+ + T(t)h_- = T(t)|h| \quad \text{a.e.}$$

The difference is positive a.e. and integrable against $u > 0$; hence

$$r\langle u, |h| \rangle < \langle u, T(t)|h| \rangle = \langle T(t)u, |h| \rangle = r\langle u, |h| \rangle,$$

a contradiction. Hence $e^{-t\mu_0} = \|T(t)\|$ is a simple eigenvalue with eigenvector $\Omega_L > 0$ a.e., t -independent by spectral calculus.

Gap. The spectrum of K_L is discrete (Theorem 5.2); μ_0 is simple; hence $\mu_1 > \mu_0$.

The state. Each $W_{\text{ct}}(\zeta)$ is unitary on $L^2(Q, \mu) \cong \mathcal{F}$; $\omega_{L,\lambda}^{\text{ct}}$ is a state on the C^* -algebra they generate. $\square$

8 Acceptance, ledger, imports

Acceptance against the scoping statement

Statement A1's four clauses are Theorems 4.1 (self-adjoint bounded-below realization; operator and form sum on the cylinder core), 5.2 (trace class, discreteness, free comparison, trace-norm cylinder approximation), 7.1 (simple ground state, gap, $\omega_{L,\lambda}^{\text{ct}}$), and 6.2 (periodic FK with cylinder compatibility and the $e^{-\mathcal{U}} \in L^p$ package); A2(b) is Corollary 6.3. The scoping deliverable “import audit or proof” is resolved as: *proof*, on the sanctioned fallback route, with the import decision recorded in §1.3 and the dossier images for the single printed cluster consumed.

claim	status	ground
Pointwise bound and coercivity of $V^{(K)}$; $b_K \leq c_1 \log^2$ -growth	proved	Lemma 2.2
Chaos structure; $\ V^{(K)} - V_L\ _2 \leq C \log K / \sqrt{K}$; all L^p	proved	Lemma 2.3
Abstract uniform Nelson bound (dyadic-in- K form)	proved	Lemma 2.5 (= V2 Thm. 7.2 pattern)
Uniform L^p bounds: vacuum, OU-interval, periodic (plain and shifted)	proved	Prop. 2.6, Lemma 6.1(iv)
Uniform inf spec $\geq -c_\lambda$; $K^{(K)} \geq \frac{1}{2}H_0 - c_{1/2}$	proved	Corollary 2.7

claim	status	ground
Finite-mode package (Gaussian unitary, closed form, compact resolvent, Mehler–Trotter FK kernel, cyclic bridge density, OU matrix elements, periodic trace, mean shift)	proved	Theorem 3.1 (explicit derivation; normalization dictionary in Rem. 3.2)
K_L : strong semigroup limit, C_0 , self-adjoint, $\geq -c_\lambda$; operator/form sum on $\mathcal{D}$	proved	Theorem 4.1
Hypercontractive smoothing $\ T_K(s)\ _{2 \rightarrow 4} \leq M_0 e^{c(s-s_4)}$, K -uniform	proved	Lemma 5.1
Norm Cauchy at t_2 ; eigenvalue convergence; trace class $\forall t$; $\mathcal{J}_1$ -convergence; weighted traces	proved	Theorem 5.2
Partition identity $Z_\infty = Z_0 \mathbb{E}[e^{-\mathcal{U}}]$	proved	Theorem 6.2(a)
Weyl mean-shift identity for e^{-tK_L} , all $D4$ -refined symbols	proved	Theorem 6.2(c)
A2(b)	closed	Corollary 6.3
Positivity improving; simple ground state; $\gamma_L > 0$; $\omega_{L,\lambda}^{\text{ct}}$	proved	Theorem 7.1
ESA of $(H_0 + V_L) _{\mathcal{D}}$ (Segal)	<i>not consumed</i>	scope fence, Remark 4.2

External imports

Only Import 1.1 (Simon book, Thms. I.12, I.16, I.17, Cor. I.15; the identical cluster as Part I’s A:hyper; PDF in refs/; page images `simon-gamma-p{48,53,55,56,57}.png` in `part_ii/dossier_images/`, printed pages 26/31/33/34), the elementary Trotter instance for bounded perturbations (the same invocation as model-sheet (3.16)), and the program-internal items of Import 1.2. The Høegh-Krohn (1974) and Figari–Høegh-Krohn–Nappi (1975) candidates were *not* acquired and are *not* consumed; the scoping’s fallback clause is exercised instead.

Phase V3 gate: A1(i)–(iv) proved, A2(b) closed. Remaining phase: V4 — A6 (fixed- t convergence via Vitali, consuming V2’s L^2 comparison and this document’s trace identities), A7 (partition functions and the derived eventual uniform lattice gap, consuming Theorem 6.2(a) and $\gamma_L > 0$), A8 (assembly through the Weyl criterion), and the adversarial pass over the assembled Part II chain.